

One defect class in the MACH-Aero adjoint: a branch recorded on a differentiated path the slope limiter in parallel, the slope limiter in serial, the IDWarp rotation guard — and the ILU conditioning wall beside them

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Abstract

On a 2,777-cell OpenFOAM case run under `dafoam/opt-packages:latest` (DAFoam 5.0.0, OpenFOAM v2506), varying *only* the mesh decomposition moves the analytic $dC_D/d(\text{shape})$ gradient from 0.00054% agreement with its own finite difference (`simple` 4x1x1) to 8.95% disagreement (`scotch`, the shipped default), with the converged primal invariant to five significant figures and the adjoint right-hand side invariant to 4.8×10^{-6} . The linear solve is not at fault: it converges cleanly (`PetscConvergedReason: 2`, true residual 1.7×10^{-7} in the unpreconditioned norm) and is unmoved by a tolerance tightened four decades. The discriminating measurement is a cross-residual: the converged `scotch` adjoint vector, permuted to serial ordering by integer `decomposePar` addressing and evaluated under the serial reverse-AD operator, leaves a residual of $329 \times \|b\|$ against that operator's own 1.14×10^{-4} floor — concentrated in the x -momentum rows of partition-interface cells. The parallel matrix-free transposed-Jacobian product is not the transpose Jacobian of the residual it claims to differentiate.

What carries it is now named to one word. Changing `div(phi,U)` from `bounded Gauss linearUpwind limited` to `bounded Gauss linearUpwind default` — same scheme family, same order, the same halo-exchanged gradient-correction term still evaluated, with only the limiter's min/max selection over the stencil removed — takes the gradient error from 8.95% to 0.849%, the adjoint Krylov count from 719 to 41, and the cross-residual from $329 \times \|b\|$ to $1.35 \times 10^{-2} \times \|b\|$: a 24,270-fold collapse of the operator error, recomputed from the raw dumped vectors by an independent sweep. The other lever that cleans this gradient does not clean the operator. Swapping the farfield `U` boundary condition leaves the transposed-Jacobian product wrong at $1.047 \times \|b\|$ — 163,548 times its own serial floor — underneath a `check_totals` reading 0.019%. An earlier revision of this document called that configuration clean. It is not safe; it is quiet.

The same recorded branch then broke a case with no processor boundary in it at all. Installed on a 4,032-cell structured NACA0012, the limiter drives the *serial* adjoint to 92.8% against its own step-stable finite difference, with one of eight gradient components sign-flipped — and the identical one-word edit returns 0.121%. Both decompositions of that case agree to 3.9×10^{-4} while both are 92.8% wrong: invariance certifies consistency, not correctness.

We report the phenomenon, the discriminators that convicted the operator and exonerated every other candidate, the localization, the campaign that named the branch and separated the fix from the disguise, the serial variant and its single-process reproducer, why no published verification could have caught any of it, and two adjacent findings from the same forensic arc: an IDWarp degenerate-branch guard that differentiates to a hard zero at exactly the state every gradient evaluation uses (root-caused, patched in four lines, and answering a five-year-old open upstream issue), and an ASM sub-block ILU factorization whose failure is broken by the PETSc developers' own sanctioned escalation to complete LU. **Nothing in this document has been filed upstream.**

Status. No issue has been opened, no maintainer contacted, nothing posted. This document is the extended, citable companion to two prepared-but-unfiled reports, `demo-output/website/dafoam/UPSTREAM_BUG_REPORT_decomposition_adjoint.md` and `demo-output/website/dafoam/UPSTREAM_BUG_REPORT_mesh_warpDeriv.md`. It does not replace them. Both remain **PREPARED**: complete, carrying the 2026-08-07 addenda these sections summarise, and *held*. Submissions are parked at the owner’s discretion; whether any of this is sent upstream is the lab principal’s call, not the authors’. Where our own record was found defective by our own verification, the defect is printed here rather than repaired silently.

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1 Summary of findings

Four findings, from four weeks of forensic work on the MACH-Aero/DAFoam toolchain. Three of them are one defect class — a *branch* recorded on a differentiated path, whose selection the reverse sweep gets wrong — in two repositories, in parallel and in serial. The fourth is a wall in a shipped default with a remedy that is one option away.

What links the first three. A guard, a limiter, and a switching boundary condition are the same kind of object: a `min`, a `max`, an `abs`, or an `if` evaluated on data and then recorded on a tape. Reverse-mode AD differentiates the branch that was taken, correctly and by design. What it cannot do is know that the branch it recorded is the wrong function of the inputs — a constant where the true map has a first-order term (D2), or a selection whose stencil reaches across a processor face and whose reverse does not (D1), or one that is simply mis-swept in a single process (D1s). The class is cheap to hunt once it is named: every member here is removable by a one-line dictionary edit or a four-line derivative, which makes it the cheapest discriminator available, and it was one such edit that closed D1’s mechanism hunt.

What links all four. They are invisible to the checks that ship with the tools. D2’s discarded term sums to *exactly* zero over the surface, and the sum is the only quantity upstream’s test asserts on. D1 produces a gradient that is self-consistent under the decomposition it was measured on, and no published verification varies the decomposition. D1s is self-consistent across decompositions too — it passes the invariance gate this lab adopted after D1, to 3.9×10^{-4} , while being 92.8% wrong. D3 announces itself loudly, but the documented remedy ladder does not contain the remedy. Each was found by measuring the thing the instrument was not looking at.

2 The decomposition-dependent adjoint

2.1 The phenomenon

The case is A4, a coarse Ahmed body: 2,777 cells, snappyHexMesh-class grid with refinement interfaces, DASSimpleFoam, kOmegaSST, one FFD shape design variable. The instrument is OpenMDAO’s `check_totals`, whose finite-difference column is computed *per run, inside that run’s own check_totals* — not computed once and reused. Only the decomposition varies across the rows of

Finding	Where	Headline measurement	Status
<i>The class: a branch recorded on a differentiated path.</i>			
D1. The limiter, in parallel: a decomposition-dependent adjoint	DAFoam v5, matrix-free (Jacobian-free) adjoint; <code>cellLimited Gauss linear 1</code> feeding <code>linearUpwind</code> , recorded on the reverse tape across processor boundaries	Gradient error 8.95% at np=4 scotch vs 0.00054% at np=4 simple 4x1x1 on the same mesh, same script, same container. Cross-residual of the converged scotch adjoint under the serial operator: 3.288×10^2 of $\ b\ $, against a 1.14×10^{-4} serial floor. One word of fvSchemes (limited $\rightarrow$ default) takes those to 0.849% and 1.35×10^{-2} , and the Krylov count from 719 to 41	Mechanism named to a recorded construct, not to a source line. Report drafted with two addenda, not filed . No prior report of the decomposition phenomenon anywhere (63 searches, 10 venues); the limiter result post-dates that sweep and has had none
D1s. The same limiter, in serial	Same <code>cellLimited</code> feeding <code>linearUpwind</code> , on A1's 4,032-cell structured NACA0012 at np=1 — no processor boundary anywhere in the case	$dC_D/d(\text{shape})$ 92.8% against its own finite difference, which moves 2.10% between $h = 10^{-3}$ and 3×10^{-3} — 44x too little to explain the gap; C_L/shape 7.9%; one of eight components sign-flipped, with both one-sided FD slopes on the wrong side of zero from the analytic. np=1 and np=4 agree to 3.9×10^{-4} while both are 92.8% wrong. The same one-word edit reads 0.121%	Decomposition-independent — the first in this class. Single-process reproducer, one-word on/off. Carried into the D1 report as its 2026-08-07 addendum, not filed
D2. Degenerate-branch rotation guard	IDWarp 2.6.2, <code>src/utils/vectorUtils.f90:58</code> , reverse mode <code>vectorUtils_b.f90:123-128</code>	<code>warpDeriv</code> discards the surface-normal-rotation term exactly whenever current normals equal reference normals — the state of every gradient evaluation. 210%/213% (upstream's own test), 634% sign flip, 207%/122% sign flips; all collapse to FD-truncation level under a four-line analytic-limit patch, primal md5-identical	Root cause confirmed by repair. Matches open upstream issue <code>mdolab/idwarp#57</code> (2021-07-14, zero comments). Comment-on- <code>#57</code> prepared, not filed
<i>Adjacent: a shipped default whose own remedy ladder does not contain its remedy.</i>			
D3. ILU conditioning wall	DAFoam v5, <code>DALinearEqn.C: 266-267</code> : ASM sub-block PC hard-coded to PCILU	<code>PetscConvergedReason: -9</code> (<code>DIVERGED_NANORINF</code>) at iteration 0 on a 21,000-cell closure case becomes reason 2 in 667 iterations under complete LU in the sub-blocks; the resulting 21,000-component field-inversion gradient verifies against central differences at 0.085%, 0.059%, 0.199%	Fixed locally by a one-hunk, default-off source patch. PETSc-sanctioned remedy; PR-ready in principle, not filed

Table 1: The four findings. Sources: Sections 2, 3, 5, 6 respectively.

Table 2; verified by `diff` of every run directory against the source case (the `np` runs carry a byte-identical `runScript.py`; the `simple` runs differ by exactly one inserted `decomposeParDict` `daOption` line; the tolerance run by exactly the `gmresRelTol` value), and by reading the `Decomposition method` line back out of each log.

configuration	log	analytic	FD (same run)	rel. err	
np=1, stock	a4_np1_stock.log	2.3965e-01	2.4232e-01	1.10%	
np=1, patched IDWarp	a4_np1_patched.log	2.4150e-01	2.4232e-01	0.34%	
np=2 scotch	a4_np2_patched.log	2.4118e-01	2.4182e-01	0.26%	
np=3 scotch	a4_np3_patched.log	2.5641e-01	2.4178e-01	6.05%	analytic high
np=4 scotch	a4_np4_patched.log	2.2086e-01	2.4258e-01	8.95%	analytic low
np=4 scotch, gmresRelTol 1e-10	a4_np4_ tol1.0e-10.log	2.2086e-01	2.4258e-01	8.95%	811 iters, reason 2
np=4 simple 4x1x1	a4_np4_ simple4x1x1.log	2.4220e-01	2.4220e-01	0.00054%	
np=4 simple 1x4x1	a4_np4_ simple1x4x1.log	2.4379e-01	2.4265e-01	0.47%	
np=4 simple 4x1x1, stock	a4_np4_simple411_ stock.log	2.4037e-01	2.4220e-01	0.76%	
np=4 scotch, stock	a4_h1e-3_stock.log	2.1821e-01	2.4258e-01	10.04%	= published

Table 2: The verified phenomenon: $dC_D/d(\text{shape})$, analytic against each run’s own finite difference. Every cell re-extracted from the run logs by an adversarial supervisor sweep, `VERIFICATION_A4_decomposition_supervisor_sweep.md` (27d25762), Axis 1. Logs under `/home/ubuntu/certonomous-runs/W4-a4-stepsweep/`. “patched” denotes a locally patched IDWarp removing the independent defect of Section 5; the decomposition effect exists stock too, as the last two rows show.

Four properties of this table decide everything that follows.

1. **The primal does not move.** Converged baseline C_D spans 0.15296979–0.15297237 across *all* configurations — invariant to five significant figures. `np=1` and `np=4 scotch` differ by 5.5×10^{-7} absolute, 3.6×10^{-6} relative: separately reconverged SIMPLE solves at `primalMinResTol` 1e-4, i.e. reconvergence noise, four orders of magnitude below the 8.95% gradient shift. The `simple` arm — the arm with the *correct* gradient — differs from `np=1` by *more* than `scotch` does.
2. **The finite difference does not move meaningfully.** The FD column spans (2.4178–2.4265)e-01, a 0.36% spread, against an 8.95% analytic excursion.
3. **Rank count is not the trigger.** `np=2 scotch` sits at the `np=1` floor (0.26%); `np=3` is catastrophic in one direction (6.05%, analytic high) and `np=4` in the other (8.95%, analytic low); `np=4 simple 4x1x1` is the cleanest configuration in the table. What changes between the dirty and clean rows is the *shape and orientation of the cut*.
4. **The solve is not the problem.** Tightening `gmresRelTol` from 1e-6 to 1e-10 costs 811 GMRES iterations instead of 719 and changes the answer by nothing. The `np=4 scotch` solve reports `PetscConvergedReason: 2` at final true residual 1.749572×10^{-7} , and DA Foam’s own solver configuration makes that literal: `DALinearEqn.C` sets `KSPSetPCSide(PC_RIGHT)` and `KSPSetNormType(KSP_NORM_UNPRECONDITIONED)`, so “converged in the unpreconditioned norm” means what it says.

This is the signature of a *converged wrong answer*. It is also the signature that makes the defect survivable in production: nothing in the solver’s own telemetry is anomalous.

2.2 The discriminators

Four candidate carriers were named in advance and executed cheapest-first (DISCRIMINATORS_A4_decomposition_mechanism.md; run artifacts under /home/ubuntu/certonomous-runs/W4-a4-discriminators/, with per-arm directories, ledger.txt, runScript_w4.py, build_maps.py, analyze_mats.py).

M3(a) — the primal, exonerated at zero cost. Point 1 above, plus this session’s own arms: baseline C_D 0.1529738 / 0.1529749 / 0.1529732 (np1 / scotch / simple), and mapped primal *state vectors* agreeing across decompositions at 2.0×10^{-3} / 2.8×10^{-3} relative — with the duplicated processor-face ϕ copies agreeing to 2.7×10^{-15} , machine precision. That last number does double duty: it validates the permutation used by every measurement below.

M3(b) — the mesh-derivative chain, exonerated by measurement. The dCD/dX_v capture hook (a real reverse seed on DAfoamWarper) was run at all three decompositions with the warp finite-difference directional derivative $dX_v/dshape$ computed identically in each arm:

arm	$\langle dC_D/dX_v, dX_v/ds \rangle$	framework compute_totals	agreement
np=1	2.414994844096177e-01	2.414994857859927e-01	5.7e-09
np=4 scotch	2.208588587555006e-01	2.208588593817390e-01	2.8e-09
np=4 simple 4x1x1	2.422030559126777e-01	2.422030573188107e-01	5.8e-09

The hand composition equals the framework total to 6×10^{-9} in *every* arm: the warp/DVGeo leg composes exactly at np=4 too. The warp direction itself is decomposition-invariant — $dX_v/dshape$ mapped point-by-point via pointProcAddressing matches np=1 at **0.0e+00** in both np=4 arms. What differs is the captured dCD/dX_v seed: mapped to np=1 point ordering, **scotch** differs from np=1 by **38.4%** in norm, **simple** by 3.1%. The IDWarp/DVGeo leg is exonerated; dCD/dX_v is wrong because the ψ it contracts is wrong.

M2 — coloring and preconditioner, exonerated by architecture and by measurement. Both dRdWTPC and the full assembled dRdWT were dumped in all three arms and permuted to np=1 ordering. The diffs do not correlate with the defect: full-dRdWT Frobenius diff 4.8×10^{-2} (**scotch**) against 5.4×10^{-2} for **simple** — *the FD-exact arm*; PC diff 3.7×10^{-4} against 3.7×10^{-3} . This is a recorded negative, and the reason it is uninformative is itself measured: at np=1 the assembled dRdWT applied to ψ already disagrees with the AD product by $230 \times \|b\|$. The assembled matrix is a crude finite-difference object; only the preconditioner is built from it, which is precisely why the KSP operator is matrix-free. Diff operator *actions*, not operator *entries*.

Architecturally the exoneration is stronger still: in the Jacobian-free path the coloring machinery builds only dRdWTPC, and a wrong preconditioner cannot move a right-preconditioned GMRES solution that converged in the unpreconditioned norm (DALinearEqn.C:170,313) — which this solve did.

M1 — the cross-residual, which convicts the operator. Each arm’s converged adjoint ψ was dumped and permuted to np=1 global state ordering by an *exact integer map* built from cellProcAddressing / faceProcAddressing (sign carried for flipped faces) plus DAfoam’s own AdjointIndexing dumps (writeJacobians: ["adjointIndexing"]) — no coordinate matching anywhere in the construction. The true residual $\|A^T\psi + b\|$ was then evaluated under the *serial* AD operator via calcJacTVecProduct.

Three decisive facts follow.

1. **The scotch ψ does not remotely satisfy the serial system** — $329 \times \|b\|$, 3.3 million times the np=1 floor — while it satisfies its *own* scotch-evaluated operator, same instrument,

ψ from	$\ A^T\psi + b\ $ under np=1 operator	ratio to $\ b\ = 0.1826$	$\ \psi\ $
np=1 (control)	2.083e-05	1.14e-04	3.277e-02
np=4 scotch	6.004e+01	3.288e+02	5.411e-01
np=4 simple 4x1x1	1.711e-02	9.37e-02	1.814e-02

Table 3: The cross-residual. See Section 2.4 for the provenance of these values, which is not what the original record’s caption claimed.

same API, at 1.2×10^{-3} relative, and the KSP that produced it reports final true residual 1.7×10^{-7} (719 iterations, reason 2). Same vector, two operators, answers five orders apart: **the operators differ.**

2. **The right-hand side is not the carrier.** $b = dC_D/dW$ mapped across decompositions is invariant to 4.8×10^{-6} (**scotch**) and 6.9×10^{-6} (**simple**) relative.
3. **$\|\psi\|$ is inflated $16.5\times$ under **scotch**** (0.541 against 0.033 np=1 and 0.018 simple) at identical $\|b\|$. The extra content sits in the pressure (0.52 against 0.031) and z -momentum (0.117 against 0.005) adjoint blocks.

Confounds closed. (a) *Linearization state.* The cross-residual was re-evaluated with the np=1 operator linearized at the **scotch** arm’s own mapped state: **scotch** reads 6.004094×10^1 , unchanged to six digits, while the np=1 control degrades only from 1.14×10^{-4} to 1.0×10^{-3} . The state confound therefore contributes at the 10^{-3} level, five orders below the signal — and the **simple** arm’s state differs *more* from np=1 than **scotch**’s does (2.8×10^{-3} against 2.0×10^{-3}) while its residual is 3,500 times smaller. (b) *Duplicated- ϕ averaging in the ψ map.* Replacing the average by either copy moves the residual by at most 9.1×10^{-3} , four orders below the signal. This bound was computed with the assembled matrix — the instrument M2 discredits — which the original record did not disclose; the supervisor sweep found it and reproduced the number exactly as $\|A_{\text{asm}}(\psi_{\text{copy}} - \psi_{\text{avg}})\|$. It is immaterial by margin: even inflated by the full $230\times$ assembled-vs-AD discrepancy the bound stays roughly $30\times$ below the 60.04 signal, and the duplicate copies disagree by at most 3.6×10^{-4} to begin with. The confound stays closed; the instrument should have been flagged.

2.3 The objection this measurement must survive

A knowledgeable reader will ask: is the **scotch** operator perhaps the *exact* adjoint of a different-but-equivalent parallel discretization? If decomposition changed the discrete residual itself at processor boundaries, the **scotch** adjoint could be consistent with *its* residual while differing from serial, and its 1.2×10^{-3} self-consistency would even fit that story.

It does not survive the finite-difference column. The FD in Table 2 is computed per run, inside each arm’s own **check_totals**. An adjoint that were the exact adjoint of an equivalent parallel discretization would match the finite difference of *that* discretization. The **scotch** analytic 2.2086e-01 disagrees by 8.95% with its own decomposition’s FD of 2.4258e-01. Combined with the fact that OpenFOAM processor patches evaluate the same interior face fluxes — no legitimate $O(1)$ residual-function difference is available — and that b maps invariantly at 4.8×10^{-6} , the supported conclusion is the strong one: **the parallel operator is not the transpose Jacobian of the residual it claims to differentiate.**

This gap was not acknowledged in the original mechanism record; the supervisor sweep found it and closed it from the prior record (VERIFICATION_A4_mechanism_supervisor_sweep.md, 8E0A08BC, Axis 4). It is stated here in the body, not in a footnote, because it is the first objection a maintainer will raise.

2.4 Sign-convention provenance, stated against ourselves

The values in Table 3 appear in no log, and the original record’s caption cited a log that contradicts them on its face. The instrument (`runScript_w4.py`, tasks `w4_dump` / `w4_crossres`) computed `res = Atpsi - b`, whereas the system OpenMDAO seeds is $A^T\psi = -b$. (*Past tense as of 2026-08-07: the instrument has since been repaired — see the note closing this subsection. Every number in this document was measured under the convention described here.*) Every arm’s log therefore prints the sign-degenerate `ratio=2.000000e+00` for its own-operator check, and none of the record’s own-operator numbers — the 1.14×10^{-4} floor, the 1.2×10^{-3} self-consistency, the 2.083×10^{-5} — appears in any log.

The tabulated values are exact offline sign corrections, `res + 2b`, recomputed from the dumped vectors (`w4x_res*.numpy`, `w4_Atpsi`, `w4_b`). The correction is exact, and the adversarial sweep reproduces every digit of it three independent ways:

- **Recomputation from the dumps** with the sweep’s own scripts: `np=1` 1.140697×10^{-4} , `scotch` 3.288078×10^2 (absolute 6.004094×10^1 , max entry 42.25), `simple` 9.367397×10^{-2} .
- **An independently rebuilt permutation**, constructed from `cellProcAddressing` / `faceProcAddressing` and the `AdjointIndexing` dumps by the sweep’s own parsers, bit-for-bit identical to the session’s map in `tgt` and `sgn` — and validated on a channel the original session never used: the `AdjointIndexing` dumps carry cell and face centre coordinates, and mapped `np=4` coordinates match `np=1` coordinates at **max** $|\Delta xyz| = 0.0$ over all 26,149 (`scotch`) and 26,227 (`simple`) entries. Every entry of the map is thereby validated exactly, independent of any solved field.
- **A fresh serial container run**: a new case directory staged from the same source, fresh serial primal, `w4_crossres` applied to ψ vectors mapped by the sweep’s own permutation. Raw `scotch` ratio 3.288139×10^2 ; sign-corrected 3.288078×10^2 ; `np=1` control 1.140697×10^{-4} ; `simple` 9.367397×10^{-2} (`w4-a4-verify-audit/audit_crossres.log`, 9 s, 0.30 core-min).

The defensible citation for Table 3 is therefore the dumped vectors plus the stated sign correction — not `d_crossres.log`. Anyone reproducing this protocol should fix the convention in the instrument rather than offline; `w4_crossres2`, written later, has the sign right and its log matches its numbers.

Note added 2026-08-07: the instrument has been repaired, and the repair confirms the corrections. This document’s own instruction — fix the convention in the instrument rather than offline — had not been executed, and a family supervision pass filed that as a standing high-severity finding: the two reused tasks still carried the wrong sign, and `w4x_res*.numpy` (needing $+2b$) sat beside `w4x2_res*.numpy` (already the true residual) under same-family filenames with nothing declaring which. Both sites now compute `res = Atpsi + b`; every W4D/crossres log line prints an explicit convention banner; and the sign-fixed task writes a distinct `w4xf_res*.numpy` prefix so pre-fix and post-fix dumps cannot be conflated (`SUPERVISOR_FAMILY_REVIEW_2026-08-07.md`, item A-1, OUTCOME FIXED; sibling repo-side commit 9796f90f).

The repair doubles as a fourth independent reproduction of the correction, and it is the cleanest of the four because it removes the offline arithmetic entirely. Run on the existing `np=1` dump with **no offline correction at all**: the old convention prints the degenerate $\|A^T\psi - b\|/\|b\| = 2.000000$, the new convention prints $\|A^T\psi + b\|/\|b\| = 1.140697 \times 10^{-4}$ — equal to *every digit* of the published offline `res + 2b` value of Table 3’s floor. The offline corrections are therefore no longer only exact by derivation and reproducible by re-derivation; they are independently confirmed by a tool that never performs them. No published number moves, and none was re-derived from a post-fix output.

2.5 Localization

The 60.04 cross-residual is not spread over the domain. By state group, **59.78 of the 60.04 norm sits in U_0 (x -momentum) rows**. By cell, entries with $|r| > 0.5$ number 15, of which **13 sit on scotch partition-interface cells and 2 at graph distance one** — none farther. All 15 sit on `cellLevel 0` cells, away from the mesh’s refinement interfaces. The two worst (42.25 and 42.24) sit at $y = -0.1947$ and $y = +0.1949$: the y -normal cut signature.

Three structural facts sharpen this.

- **It is not a many-rank corner effect.** The worst cells each touch exactly *one* foreign rank through exactly *one* processor face; cells touching two to four foreign faces carry *smaller* residuals.
- **Cut orientation, not cut jaggedness alone, sets the magnitude.** The hierarchy in Table 2 is orientation-ordered: x -normal planar slabs 0.00054%, y -normal planar slabs 0.47%, `scotch`’s jagged mixed-orientation boundary 8.95% — and the worst cross-residual entries sit on y -normal processor faces.
- **The same instrument on the clean arm is five thousand times quieter.** Applied to the `simple 4x1x1  $\psi$` , the serial operator maxes at **0.008439** on that arm’s own interface cells (9 of its top 10 entries sit on them) — $5,006\times$ below `scotch`’s 42.25. Had `simple`’s residual also been huge, the test would have proven nothing.

The hanging-node hypothesis, refuted backwards. The natural first guess on a refinement-interface mesh is that partition cuts crossing hanging nodes are the trigger. The supervisor sweep re-classified all 7,843 internal faces itself, from `owner/neighbour/cellLevel` and the archived `cellDecomposition` maps: 456 refinement-interface faces, of which `scotch` cuts 4 (0.88%) and `simple 4x1x1` cuts 68 (14.91%). *The decomposition that cuts seventeen times more refinement interfaces is the accurate one.* The hypothesis is refuted, not merely unconfirmed.

2.6 The mechanism, named at operator level

Named. Under decomposition, DAfoam v5’s reverse-mode AD product $(\partial R/\partial W)^\top v$ — the global-tape shell operator the adjoint KSP iterates (`initializeGlobalADTape4dRdWT / dRdWT-MatVecMultFunction`, `DASolver.C:1364ff`) and the `calcJacTVecProduct` API that reverses the same recorded computation — differs from the true transposed Jacobian of the discrete residual at partition-interface cells, by amounts that depend on the cut’s shape and orientation.

The two parallel AD paths sit *together* on the wrong side: they agree with each other to 1.2×10^{-3} of $\|b\|$ while both sit 3.3×10^2 of $\|b\|$ from the serially evaluated truth. The defect is therefore common to the recorded reverse computation, not to one call site. GMRES then solves the wrong system exactly; the adjoint inherits $O(1)$ error along the directions the wrong operator mislabels (hence the $16.5\times$ norm inflation); and the gradient contraction $\psi^\top \partial R/\partial X_v$ delivers the 8.95%. Every observed symptom follows: converged wrong answer, tolerance invariance, decomposition dependence, primal/FD/RHS invariance, and the near-invariance of the lab’s conformal-mesh cases.

Not claimed, at the time this was first written. The defective source line. The candidate surface was stated as the reverse-AD handling of inter-processor halo exchange inside the recorded residual evaluation — the tape over OpenFOAM processor-boundary updates — and pinning the line requires an instrumented rebuild of `libDASolver.so`. What was pinned: the subsystem, the spatial signature (interface-cell x -momentum rows), the orientation dependence, and the discriminating instrument that any candidate fix must satisfy.

What has since been pinned. The construct the tape records. A campaign registered to attack a different weakness came back with the carrier in the case’s own `fvSchemes`: the slope limiter’s `min/max` selection over a stencil that reaches halo cells. One word removes it, and removing it collapses the operator error by four and a half orders of magnitude (Section 2.9). The line is still not pinned — but the search space has narrowed from “the tape over halo exchange” to “the reverse sweep of `cellLimited/limitedGrad`”, and it now has a *single-process* reproducer to instrument (Section 3), which is far cheaper than a parallel one.

A continuum, not a binary. The FD-exact `simple 4x1x1  $\psi$` still leaves 9.4×10^{-2} of $\|b\|$ under the serial operator — roughly $800\times$ the `np=1` floor — while its gradient is correct to 0.00054%. Every `np=4` operator tested differs measurably from serial; `scotch`’s jagged cut is the catastrophic point on a continuum, not a separate species.

2.7 A usability finding recorded on the way

The last and most expensive discriminator — run with coloring disabled — could not deliver its number, and the reason is worth reporting to upstream in its own right.

`adjUseColoring: False` crashes the **v5 mphys Krylov path by construction**: `mphys_dafoam.solve_linear` skips `runColoring()` when the option is `False`, but `calcdRdWT` unconditionally calls `readJacConColoring()`, which reads a file that was never written and aborts in `DAColoring::validateColoring`. The static chain was re-confirmed in the shipped container by the supervisor sweep: `mphys_dafoam.py:458` skips `runColoring()` unless `adjUseColoring` is `true`; `DA-Solver.C:1043` calls `readJacConColoring()` unconditionally; and `DAColoring::validateColoring` (beginning line 931) reaches its `FatalErrorIn ... abort` at approximately `DAColoring.C:1021`.

The identity-coloring branch (one colour per global column) is reachable only through `runColoring()`, so the arm was rerun with an explicit `runColoring()` call before `compute_totals`: identity coloring accepted, `nJacConColors: 26149`, preconditioner partials computed by pure per-column finite difference with no colour compression. **The forced path is memory-unbounded.** On this 2,777-cell case (colored path peaks under 2 GiB) the first forced attempt thrashed an 8 GiB container cap and was killed at 701 s; a second at a 20 GiB cap completed all 26,149 column evaluations in 258 s and then exhausted 20 GiB in the post-assembly stage — consistent with per-column FD columns of an elliptically coupled residual being nearly dense — and was killed thrashing at 591 s.

The pre-registered prediction for this arm (“the gradient will not move, because coloring builds only the preconditioner”) is scored **NOT SCORED**: the discriminator never met its measurement. The coloring exoneration does not rest on it; it rests on architecture plus M1, as set out above.

2.8 How far the defect reaches

The trigger condition is bracketed by measurement rather than asserted from one case. Four cases in this lab are already on record as decomposition-invariant — A1 (4,032-cell structured airfoil, `np=1` vs `np=4`: relative error 3.795529×10^{-4} vs 3.807681×10^{-4}), A2 (38,304-cell conformal wing), A5 (4,800-cell conformal U-bend, six 27-component gradient groups all invariant at norm level, worst group-norm difference 3.8×10^{-4}) — and every clean case is conformal while the only dirty case is the only snappy-refined mesh tested. That confound is the target of a pre-registered breadth matrix, `DEFECT_REACH_decomposition_cases.md` (pre-registration committed at 99F5D41D before any arm ran).

Status at the time of writing (2026-08-05). All arms have run — the ledger at `/home/ubuntu/certonomous-runs/W4-defect-reach/ledger.txt` records thirteen entries from the Ahmed-35 mesh generation through the contingent N7 cross-residual — but the results section of the reach record is still being written and its predictions are not yet scored. One result has been

arm	configuration	pre-registered prediction
N1	A4, np=4 simple 1x1x4 (z-normal slabs)	rel. err in [0.05%, 1.5%]: planar slabs of any orientation benign-to-moderate
N2	A4, np=4 simple 2x2x1 (4-rank corner line)	rel. err $\leq 1\%$. Decision rule: if $> 2\%$, “scotch-style jaggedness required” is REFUTED
N3	Ahmed-35, a <i>new</i> snappy-refined case built by A4’s own recipe, np=1	control: rel. err $\leq 1.5\%$
N4	Ahmed-35, np=4 scotch	headline: defect appears, rel. err $\geq 2\%$, analytic differing from N3 by $\geq 2\%$ while FD stays within $\sim 0.5\%$
N5	Ahmed-35, np=4 simple 4x1x1	clean: rel. err $\leq 1\%$ regardless of N4
N6	CBFS, 21,000-cell conformal blockMesh, 21,000 beta DVs, simple 4x1x1 vs the scotch record arm	decomposition-invariant: $\leq 1\%$ per component at the three recorded FD cells, norms to $\leq 2\%$
N7	Ahmed-35 cross-residual (contingent on N4)	$\geq 10 \times \ b\ $ under the np=1 operator, on momentum rows of interface cells

Table 4: The pre-registered reach matrix. Predictions committed before any solver ran; scoring is HELD / NOT HELD / NOT SCORED exactly as written, with no post-hoc bands.

relayed from that work in progress: **N6, CBFS, is decomposition-invariant at 1.1×10^{-4}** , consistent with the pre-registered prediction and with the conformal-mesh pattern. *The remaining arms are marked in progress here deliberately; this document will not pre-empt a pre-registered scoring with numbers read out of raw logs.*

Addendum, 2026-08-05: the reach matrix, scored and audited. The campaign has since been scored against its registered wording and adversarially audited (DEFECT_REACH_decomposition_cases.md, results appended; supervisor sweep demo-output/website/dafoam/VERIFICATION_reach_matrix_supervisor_sweep.md, 192c970c — five audit axes CONFIRMED, one CONFIRMED with a named evidential caveat). The trigger claim it confirms reads: a **freestreamVelocity** farfield U BC in the recorded tape is *necessary* for the defect, and the cut geometry modulates it (**scotch** np ≥ 3 worst, planar slabs benign). The wrong-operator defect reproduces on Ahmed-35 — cross-residual $5.45 \times \|b\|$ under the serial operator against a 3.98×10^{-4} np=1 floor, recomputed by the sweep to every printed digit under a byte-identically rebuilt map validated at max $|\Delta xyz| = 0.0$ — and a within-case separating control (N9: **inletOutlet** farfield, *no patchVelocity* input, same mesh, same np=4 **scotch** cut, registered pre-run at 15:23:54Z with 20 s of margin) lands the analytic in the clean class, print-identical to the **patchVelocity**-registered arm. Two confounds travel with the confirmation and bound the breadth claim: (i) Ahmed-35 is a deliberately controlled *sibling* of A4, not an independent geometry — same 2,777-cell background mesh family, same 25,821-state serial system, same 328-face **scotch** cut hitting the same dominant serial cells — so the defect side of the seven-case BC survey is one case family, not two independent draws; (ii) across the survey’s external-flow cases **freestreamVelocity** is perfectly anti-correlated with **patchVelocity** registration, so the cross-case survey alone cannot separate the BC from the registration. The N9 within-case swap breaks both confounds, and the causal weight of the trigger claim sits there; the survey is corroboration, not proof. This addendum supersedes the in-progress markers for the reach matrix elsewhere in this document (the artifact index and the open-items list).

Two corrections to this addendum, both from work that came after it and both printed in the sections below. First, the necessity claim survives but sufficiency does not: the same BC installed on a clean case from a different mesh family produces no defect at all (Section 2.11), and the ingredient the survey was missing turns out to be the limiter. Second, and more consequentially,

the N9 control was read here as putting the `inletOutlet` configuration “in the clean class”. That is true of its gradient and false of its operator, and the difference was measured (Section 2.10). The sentence to keep is not that `inletOutlet` is safe.

Two record facts the matrix already leans on need no new run. First, the `np=2-clean / np=3-dirty` contrast (0.26% vs 6.05%) rules out rank count as the trigger by itself. Second, the “cut through the refinement region” discriminator is already on record backwards, as set out in the localization above — and, since this paragraph was written, forwards as well, by a controlled pair (Section 2.11).

2.9 The branch, named to one word

The campaign that named the carrier was not looking for it. It was registered to attack the weakness the reach sweep had named — every defective measurement in this lab lived on one 2,777-cell Ahmed background mesh — by transplanting the trigger onto a second mesh family, by refining the first one, and by turning setup knobs that should not matter (`DEFECT_ROBUSTNESS_mesh_and_setup.md`: pre-registration 25F52868, amendments 671F40BB, C22B0F00, B0AA6102, 4EA782E3, results 49631723; run artifacts under `/home/ubuntu/certonomous-runs/W4-defect-robustness/`).

The knob arm was registered as a formality. R3a changed the discretization — convection scheme and default gradient scheme together — with the prediction that the error would stay $\geq 4\%$, because “the defect lives in the reverse tape of the parallel coupling, which every convection/gradient scheme feeds through the same processor-patch machinery.” It came back at **0.228%**, on the same mesh and the same `scotch` partition that reads 8.95%, and the registered decision rule — “error $\leq 1\%$ means the defect is scheme-specific — REOPENS the mechanism question, reported loudly” — fired. R3a had moved two knobs, so it convicted neither. Two separating arms were registered before either ran; then two sharper ones. The result is a ladder of four convection schemes — plus the gradient-scheme arm tested alongside them — in which every row keeps the same mesh, the same `np=4 scotch` cut and the same `freestreamVelocity` farfield, and is judged against its own in-run finite difference.

<code>div(phi,U)</code>	limiter	2nd	grad-corr.	analytic	FD (own run)	rel. err	KSP
<code>linearUpwind limited</code> (established)	yes	yes	yes	2.2086e-01	2.4258e-01	8.95%	719
<code>linearUpwind default</code> (R6b)	no	yes	yes	2.0572e-01	2.0399e-01	0.849%	41
<code>Gauss linear</code> (R6a)	no	yes	no	2.0299e-01	2.0267e-01	0.157%	223
<code>upwind</code> (R3a1)	no	no	no	2.7681e-01	2.7670e-01	0.041%	41
<code>gradSchemes default</code> <code>limited only</code> (R3a2)	yes	yes	yes	3.1706e-01	3.2562e-01	2.63%	780

Table 5: The scheme ladder. “limiter” is the min/max selection over the stencil; “grad-corr.” is the face correction that needs a halo exchange of neighbour values in the forward sweep and its transpose in the reverse. Every row’s error is its own analytic against its own run’s finite difference; the discretizations differ, so the analytic columns legitimately differ too. R6b is the load-bearing row. Every number re-extracted from the logs by the supervisor sweep. The last row keeps the convection limiter and limits the *default* gradient scheme instead: it modulates the magnitude by 3.4x and cures nothing.

R6b is one word, and the sweep checked that literally. `limited` names A4’s `cellLimited Gauss linear 1` gradient scheme; `default` names its unlimited `Gauss linear`. A diff of the two case directories shows exactly one changed line, in `system/fvSchemes`, with `fvSolution`, `controlDict`, `decomposeParDict`, `blockMeshDict`, `snappyHexMeshDict`, the whole of `0.orig` and `runScript.py` all byte-identical. Same scheme family, same order, the same gradient-correction term still evaluated and still requiring its halo exchange. **The only thing removed is the selection.**

The registration named the outcome that would have refuted the hypothesis, in advance: “if R6b is DIRTY ($\geq 4\%$) while R3a1 was clean, the limiter branch is NOT the carrier — the second-order gradient-correction TERM is, branch or no branch, and the supervisor’s unifying hypothesis is REFUTED for this defect and must be reported as refuted.” R6b came back at 0.849%. Read across the rows, three candidates die by a row that keeps everything else: the scheme’s *order* is not it (R6a is second order and clean), the halo-exchanged *gradient correction* is not it (R6b keeps it and is clean), and the default gradient scheme is not the gate (R3a2 keeps the limiter and stays dirty at 64x the clean floor). And the conditioning travels with the same word: **719 Krylov iterations to 41, a 17.5x collapse**, on an operator whose only change is that a min/max left the tape.

The control that refuses to call it a complete cure. A registered follow-up ran the identical unlimited discretization at np=1, predicting the surviving 0.849% would prove to be that discretization’s own protocol floor. It is not: np=1 reads **0.016%**, the prediction is NOT HELD, and a real decomposition effect at $\sim 0.8\%$ survives the limiter removal. Reading the two rows together locates it, and the location matters: the *analytic* is 2.0572e-01 at both np=1 and np=4 **scotch** — identical at all five printed digits — while the FD column moves 2.0568e-01 $\rightarrow$ 2.0399e-01. With the limiter gone the adjoint itself is decomposition-invariant at printed precision, and the remainder is an FD-column shift, not an analytic error.

The erratum, which runs in our favour. The adversarial sweep of this campaign found exactly one factual error anywhere in it, and it was ours: the established **scotch** np=4 Krylov count had been written **590** in six places. A fleet-wide grep places “590 iterations” in exactly one log — `a4_simple2x2x1.log`, a *different* arm, the **simple** 2x2x1 partition at 1.40% error. Every log that ran the established arm reads **719**, fifteen of them. Corrected at every site; the correction enlarges the claim, because the conditioning collapse is $719 \rightarrow 41 = \mathbf{17.5x}$, not the 14.4x that was reported (VERIFICATION_defect_robustness_supervisor_sweep.md, 8BD47B35: six axes, five CONFIRMED, one CONFIRMED WITH A NAMED ERRATUM).

2.10 Operator or contraction: the two levers are not the same kind of fix

By this point two edits were known to clean this gradient: the farfield BC swap (`freestreamVelocity` $\rightarrow$ `inletOutlet`, the reach matrix’s N9 control) and the limiter removal. Both had been measured only with `check_totals`. This lab’s own L-36 is the standing warning that a passing gradient check certifies a contraction, not an operator — so R5 was registered to ask which kind of fix each lever is, with the loud alternative written out in advance: “*if either lever leaves the cross-residual $\geq 50 \times \|b\|$ while its gradient is clean, then that lever does NOT fix the operator — it only rotates the objective’s adjoint direction away from the corrupted subspace, exactly L-36’s failure mode, the operator is still wrong for every other objective and DV on that case, and the upstream report’s framing must change from ‘these configurations are safe’ to ‘these configurations hide it.’*”

The instrument is Section 2.4’s cross-residual, unmodified, with each lever’s own configuration supplying its own serial operator and its own floor.

The scheme lever fixes the operator. $328.81 \rightarrow 0.013548$ is a collapse by a factor of **24,270**. The np=4 analytic also matches np=1 to 2.5×10^{-5} . Removing the limiter branch does not rotate the objective away from the damage; the damage is gone.

The BC lever hides the defect. At $1.047 \times \|b\|$ the `inletOutlet` configuration’s **scotch** adjoint fails to satisfy its own serial adjoint system by a residual the size of the right-hand side itself — **163,548 times its own np=1 floor** — underneath a `check_totals` reading 0.019%. For scale, the reach campaign convicted the Ahmed-35 case of carrying the wrong-operator defect

configuration (all np=4 <code>scotch</code> , same mesh, same cut)	gradient err	$\ A^\top \psi + b\ /\ b\ $	its own np=1 floor	ratio to floor
established (<code>freestreamVelocity</code> + <code>linearUpwind</code> limited)	8.95%	3.2881e+02	1.1407e-04	3.0×10^6
R5a — scheme lever (<code>upwind</code> , BC kept)	0.041%	1.354841e-02	1.998129e-06	6.8×10^3
R5b — BC lever (<code>inletOutlet</code> , limiter kept)	0.019%	1.047165e+00	6.402838e-06	1.6×10^5

Table 6: The two levers at operator level. Every figure recomputed by the supervisor sweep from the raw dumped vectors with its own scripts: the staged corrected residuals equal its own $\text{res} + 2b$ to the last bit, its independently rebuilt np4→np1 permutation equals the staged input exactly (`np.array_equal` true, both levers), and the sign convention was verified rather than assumed — the correction is the only reading under which the np=1 controls collapse.

at $5.45 \times \|b\|$; 1.05 is the same order of wrongness. This is L-36’s failure mode caught in the act: the gradient is clean because the contraction is small, not because the operator is right. **The configurations this lab called safe are not safe; they are quiet.** That is the most consequential sentence the campaign produced, and it is a correction to this document’s own earlier text rather than a claim about anyone else’s.

Three disclosures the sweep required, printed rather than resolved. (a) The registered band was $\leq 5 \times \|b\|$, and by that wording alone *both* levers HELD — 1.047 is a collapse from 329. The substance, that one lever hides rather than fixes, rests on the floor-ratio criterion, which was *not* a registered band. The sweep says so plainly, and says the band was mis-calibrated by roughly five orders of magnitude against the actual np=1 floors (10^{-6} to 10^{-4}). The extra-registration reading runs against the lab’s own prior framing, so nothing is smuggled — but future cross-residual bands go in units of the configuration’s own floor, not of $\|b\|$. (b) No state-override control was run for either lever, so the linearization-state confound is bounded only indirectly: by the established arm’s own precedent (328.808 with the confound removed against 328.814 without — a fourth-digit effect) and by the sibling scheme lever putting the entire protocol floor at 1.35×10^{-2} , 77x below the 1.047 signal. The conclusion survives; the ~ 0.4 core-min control is unbought and is on the open list. (c) One corroboration fell out free: the BC-lever dump prints `framework_dCDdshape=2.406182395247966e-01`, the N9 record value to all sixteen digits, from a separately staged case directory and a different task.

2.11 What the campaign eliminated, and the confound it did not

Hanging nodes, refuted forwards. Section 2’s localization refuted the refinement-interface hypothesis backwards — the accurate decomposition is the one that cuts seventeen times more refinement interfaces. R4 refuted it forwards, with the controlled pair the reach record had to list as untested: A4’s own recipe, same STL, same background `blockMesh`, same `0.orig`, same `fvSchemes` with the limiter present, and the two snappy refinement levels dropped to zero.

mesh	hanging nodes	analytic	FD (own run)	rel. err
A4 coarse (record)	502 cells at <code>cellLevel</code> 1, 456 interface faces	2.2086e-01	2.4258e-01	8.95%
A4 conformal (R4)	none (2,336 cells, <code>cellLevel</code> uniformly 0)	1.8315e-01	1.8847e-01	2.82%

Prediction $\geq 2\%$: HELD. Same geometry, refinement removed, nothing else changed, and the defect is still there at 70x the clean floor. The magnitude drop from 8.95% carries no mechanism

claim — removing refinement changes the mesh, hence the cut, hence the contraction, which the reach campaign already measured as case- and objective-specific.

Refinement: the conditioning collapse travels with the limiter. One uniform refinement of the background mesh (19,619 cells, 1,040 at `cellLevel` 1) could not be scored as registered, for a reason worth more than the arm: in the established configuration the `np=4 scotch adjoint` *does not converge at all*. GMRES exhausts `gmresMaxIters` 1000 at KSP residual 1.7188e-02 with `PetscConvergedReason: -3, solve_linear` raises, and no analytic exists to compare against anything (33.60 core-min, charged in full). R2 is scored NOT SCORED — unmeasurable as staged — and the failure is itself the datum. It rhymes with the restart arm: with the limiter present the operator is badly enough conditioned that a *restarted* GMRES stagnates on the coarse mesh (`gmresRestart` 60, stalled at 2.5079e-02 with five digits of no progress over the last hundred iterations) and an *unrestarted* one stagnates on the refined mesh, while the clean-scheme arms converge in 38–68 iterations on the same coarse mesh.

The substitute arm asked the refinement question the remaining budget could answer, and answered it: the same 19,619-cell mesh with the single R6b word removed *converges* — `PetscConvergedReason: 2` at 149 iterations, where the limited configuration exhausted 1000 — at 7.2627e-02 against its own FD 7.3195e-02, **0.777%**. That is the same scale as the coarse mesh’s unlimited 0.849%. One refinement level up, the limiter is the difference between no adjoint solution at all and a converged one, and the small surviving effect neither vanishes nor grows.

The transplant that did not fire, and what it cost. R1 mirrored the reach matrix’s N9 in reverse: where N9 removed the farfield BC from the defective case, R1 installed it on a clean one — A1’s 4,032-cell structured conformal NACA0012, a different mesh family in every respect. Registered prediction: the defect fires, $\geq 2\%$. It did not. `np=1` reads 0.043%, `np=4 scotch` reads 0.043%, and the two analytics are 1.55×10^{-7} apart. The obvious escape — “A1’s cut is too benign to excite anything” — was closed at zero solver cost with the reach campaign’s own cut classifier: A1’s `np=4 scotch` cut is **131 internal faces of which 120 are oblique**, no axis-aligned normal to 0.999, which is the maximally jagged mixed-orientation family the reach matrix identified as catastrophic; A4’s own cut is 328 faces (143 x / 7 y / 159 z / 19 oblique). And the transplant did land: baseline C_D moved 0.52% from the record configuration, the same order as the 0.44% shift N9 measured going the other way. **freestreamVelocity is necessary but not sufficient, now by measurement rather than by the absence of a test.**

The survey that retro-explains the null, and the two correlates it cannot rank. A zero-cost read of all seven cases’ on-disk `system/fvSchemes` shows why R1 was always going to be clean: A1’s convection gradient is *unlimited*, so the arm was missing an ingredient. The limiter separates the seven cases exactly as the BC does — both defective cases carry `linearUpwind` limited with `cellLimited Gauss linear 1`; all five clean cases carry an unlimited `linearUpwind/linearUpwindV`. So does a third variable: `primalMinResTol` is 1e-4 on both defective cases and 1e-6 or 1e-8 on all five clean ones. *Three perfect correlates, one seven-case survey, no ranking available from it.* What breaks the symmetry is the within-case controls, and they exist for two of the three: N9 swaps the BC alone on A4 and cleans the gradient; R6b removes the limiter alone on A4 and cleans the *operator* (Table 6). The primal-tolerance lever has no within-case control at all, because DAFoam treats an unreached `primalMinResTol` as a primal *failure* rather than a depth setting — the registered R3b arm raises `AnalysisError: Primal solution failed!` before any adjoint runs. The knob is a gate, not a control. **That confound stays open and is recorded as open.** One candidate is eliminated outright: the turbulence model is not a correlate — `kOmegaSST` sits on both sides, defective on A4 and Ahmed-35, clean on CBFS.

The limit of what the campaign establishes. Every arm that moved the defect is a within-case control on A4: one mesh, one geometry. The cross-case survey corroborates and is itself a confounded correlate. The one experiment that would have broken the one-family confound — both ingredients installed on a clean case from another mesh family — was named in the record as the next arm and deliberately not improvised, because it was in no registered edit set and the budget did not cover its full protocol. It was registered and run two days later. It is the subject of the next section, and it did not do what either registered branch expected.

3 The same branch, in serial

The outstanding experiment was named in the record before it was approved: install *both* measured ingredients — the freestream farfield BC and the limited convection scheme — on the clean structured NACA0012, and measure whether the case *acquires* the defect. It was registered as such, with its edit set, its bands, and its escape clauses (Amendment 5, 0263D950, committed seven seconds before the first arm launched), and run on 2026-08-07. The registration named two outcomes: acquisition, which would give the defect a second mesh family, or a null, which would confirm the one-family confound and force the upstream report to scope its reproducer. It produced a third.

3.1 The arm

A1 + <code>freestreamVelocity</code> + limiter	C_D /shape analytic	FD (own run)	rel. err	C_L /shape	adjoint KSP
np=1 (<code>a1lim_np1.log</code>)	1.066449e-01	6.194868e-02	92.8%	7.86%	95/96, reason 2
np=4 <code>scotch</code> (<code>a1lim_ np4scotch.log</code>)	1.066142e-01	6.194874e-02	92.8%	7.87%	96/97, reason 2

Table 7: The acquisition arm. Both arms `rc=0`; every one of the seventeen primal solves per log (baseline plus eight design variables at $\pm h$) reached the case’s own `primalMinResTol` $1\text{e-}8$, worst final residual 9.99×10^{-9} , none exiting on the iteration cap. Staging asserted in-driver; the `np=4` log reads `Decomposition method scotch` [4].

The registration’s clause (i) was a *control*: the `np=1` arm should read $\leq 0.5\%$, as its unlimited sibling did (0.043%). It reads 92.8%. The registered escape clause therefore fired — “if the `np=1` control itself is dirty ... the arm is reported unmeasurable, not scored” — and R7 is NOT SCORED against its FD-relative bands; clause (ii) is unevaluable as worded. Two facts then have to be pulled apart, and the amendment that does it (`AA7FCDDC`) was committed with the two numbers in hand and before either separating arm ran, which is stated here rather than glossed.

The acquisition question, answered: no. On the decomposition-invariance instrument the two rows are the same run twice. The analytics are 3.9×10^{-4} apart in vector-relative terms (per-component max 0.197%, median 0.040%); the FD columns 4.3×10^{-6} apart, 1.9×10^{-7} in the worst single component; the C_L analytics 3.8×10^{-5} ; the Krylov counts 95/96 against 96/97 — *no conditioning excitation at all*, where A4’s defective configuration reads 719 against its clean arms’ 41. Whatever is wrong here is wrong identically at both decompositions. **The defect did not acquire on the second mesh family.** The trigger conjunction — limiter $\times$ `freestreamVelocity` $\times$ jagged `scotch` cut — is measured *insufficient* off the Ahmed background family, and the one-family confound on the defect side stands. The remaining separating candidates sharpen rather than disappear: the Ahmed mesh family or its cut topology, or the loose `primalMinResTol`. A1 runs $1\text{e-}8$, and with both ingredients installed at $1\text{e-}8$ it is decomposition-clean — so the loose-primal co-ingredient hypothesis *gains* weight here, where a fired defect would have killed it.

One caveat is owed and the supervisor sweep required it be stated as a word rather than a hedge: **the invariance instrument on which this verdict is scored was adopted after the numbers were in hand.** It echoes a subclause the R1 registration already used, and no registered band could read a 3.9×10^{-4} shift as acquisition, so the inference is forced — but the instrument choice is post-hoc relative to the R7 registration, and this document says so.

3.2 Convicting the analytic, not the finite difference

arm	edit vs <code>a1lim_np1</code>	analytic	FD (own run)	rel. err
R7f1 <code>a1lim_np1_h3e3</code>	<code>check_totals</code> step 1e-3 $\rightarrow$ 3e-3	1.066449e-01	6.216543e-02	93.7%
R7f2 <code>a1limdef_np1</code>	<code>linearUpwind</code> limited $\rightarrow$... default (one word)	6.516374e-02	6.522800e-02	0.121%

Table 8: The separating arms, both registered before either ran. R7f1’s analytic is byte-identical to the 1e-3 arm’s at every printed digit — only the FD column moves. The sweep’s byte audit: R7f1’s only changed input is one line of `runScript.py`; R7f2’s is one word of `system/fvSchemes`.

The step check was registered to convict the finite difference: the prediction was that the FD would *not* be step-stable, the limiter having kinked the shape response at the FD scale, and that the 93% would be an unresolved-FD artifact rather than a conviction of the serial AD. It came back at 2.10% vector-relative (per-component max 15.14%), scored NOT HELD in the middle band by a tenth of a point — and read in substance rather than spun, a 2.1% step wobble is **44x too small** to account for a 92.8% gap, the FD magnitude is 6.2e-02 at both steps, and the analytic is byte-identical across them. The FD reference is essentially resolved; the discrepancy lives in the analytic.

R7f2 then puts one word between 0.121% and 92.8%, *at np=1*: the same `limited  $\rightarrow$  default` lever that cleaned the operator on A4, now convicting the limiter branch in serial on the second mesh family (C_L 0.018%, KSP 32). Everything else the staging touched is exonerated by the same arm, because R7f2 carries it all: the `linearUpwindV  $\rightarrow$  linearUpwind` family swap, the added `gradSchemes` entry, and the R1 boundary-condition edit set (0.043% on record).

3.3 The attack that could have killed it

There is a classic finite-difference trap for a branchy scheme, and it had to be closed. If the limiter’s selection flips between the $+h$ and $-h$ legs, a central difference measures a kink average rather than a slope — and a *symmetric* kink at the base point is step-invariant under central differencing, so the two-step check above cannot exclude it by itself. The record registered the kink hypothesis and then tested it only by step stability. The adversarial sweep named that gap and closed it at zero solver cost, from data the record’s own logs already carried: each FD leg prints its converged C_D , so the *one-sided* slopes were free all along.

Three readings, and the finding survives all three.

- **On the components carrying most of the gap there is no kink to blame.** For 0, 1, 4 and 5 the two one-sided slopes agree to within 10%: the function is locally smooth at the FD scale, and the analytic still misses by 65–285%.
- **On the one component that *is* branchy, the kink cannot rescue the adjoint.** Component 6 — the sign-flipped one — shows 27.4% asymmetry at $h = 10^{-3}$, growing to 91% at 3×10^{-3} : real limiter chatter. But *both* one-sided slopes are positive at *both* steps, all four secants falling between +0.0053 and +0.0143, while the analytic reads -0.025912 . The analytic lies outside the [backward, forward] bracket, on the wrong side of zero, at three times the magnitude. No branch selection available to the finite difference produces a negative slope there.

component	central FD	forward	backward	fwd/bwd asym.	analytic
0	-0.009054	-0.008926	-0.009182	2.8%	-0.034900
1	-0.020342	-0.019338	-0.021345	9.9%	-0.037237
4	+0.037311	+0.037465	+0.037158	0.8%	+0.061443
5	+0.040489	+0.040656	+0.040322	0.8%	+0.064415
6	+0.008497	+0.009660	+0.007334	27.4%	-0.025912

Table 9: One-sided slopes at $h = 10^{-3}$, reconstructed from the FD legs’ own printed converged C_D values. The same reconstruction rebuilds the log’s printed central-difference vector to 4.7×10^{-9} absolute at both step sizes, which validates the leg ordering, the step, and the framework’s arithmetic before anything is inferred from it.

- **Triangulation, also free.** The limited case’s FD vector sits at cosine **0.994** to the *unlimited* case’s analytic — which R7f2 validated against its own FD at 0.121% — while the limited analytic sits at cosine 0.90 to its own FD. Three of the four instruments agree with each other; the limited serial adjoint is alone.

The sweep’s recommendation is adopted here as protocol, because it is strictly stronger than a second step size and costs nothing: for this symptom class, reconstruct the one-sided slopes from the FD legs’ printed objective values. It belongs in the registration, not in the post-mortem.

3.4 What the serial variant is, and what it changes

- **It is decomposition-independent, and it broke a family property before anyone had written that property down.** Every defect this lab had traced in this toolchain vanished at np=1. This one does not.
- **It is the same object as D1.** `cellLimited`’s min/max stencil selection, recorded on the reverse tape, wrong through a different excitation: a parallel inconsistency on the Ahmed family, a serial contraction error on the airfoil. On A4 the identical token pair reads only 0.34% at np=1 — the serial excitation is *case-dependent*, which is exactly what a wrongly-differentiated *data-dependent* branch would produce, since different meshes activate different branches at the converged state.
- **It is far cheaper to file.** A maintainer needs one process, one 4,032-cell structured case, and one word of `fvSchemes`. No MPI in the loop, no partitioner, no mapping instrument, no cross-residual protocol. The instrumented-rebuild search space is unchanged — the `cellLimited/limitedGrad` reverse sweep — and now has a single-process reproducer to instrument.
- **It costs the campaign’s framing a clause.** “The branch defect is a parallel inconsistency” is now measured to have an exception: the branch can be wrong serially, with the parallel machinery innocent.
- **It cost this lab an instrument’s warrant.** The decomposition-invariance gate adopted after A4 passes here to 3.9×10^{-4} on a case that is 92.8% wrong. *Invariance instruments detect inconsistency between two evaluations of the same tape; a tape that is wrong identically in both sails through every one of them.* Never promote an invariance pass to a correctness verdict unless at least one side is anchored to an external reference — an own-run finite difference with a step check, or an independently evaluated operator.

4 Why it was never caught

Two independent bodies of evidence answer this: the published verification protocols of the toolchain’s own method papers, and the community record.

4.1 The papers’ own protocols

Three method papers were read in full and their verification protocols tabulated (DAFOAM_PAPERS_VERIFICATION_PROTOCOLS.md, BF6AC53B; reading notes with per-claim source tables under `demo-output/website/campaign/`).

- He, Mader, Martins, Maki, “DAFoam: An Open-Source Adjoint Framework for Multidisciplinary Design Optimization with OpenFOAM,” *AIAA Journal*, doi:10.2514/1.J058853.
- He, Mader, Martins, Maki, “An aerodynamic design optimization framework using a discrete adjoint approach with OpenFOAM,” *Computers & Fluids* 168 (2018) 285–303, doi:10.1016/j.compfluid.2018.04.012.
- Kenway, Mader, He, Martins, “Effective Adjoint Approaches for Computational Fluid Dynamics,” *Progress in Aerospace Sciences*, doi:10.1016/j.paerosci.2019.05.002.

paper	adjoint operator verified	case / mesh class	cells	decomposition	result
C&F 2018	v1 explicit FD-coloring Jacobian	Ahmed body 25°, snappy-HexMesh	~1M	unstated	avg <0.1%, worst 0.252%
AIAA J 2020	v1 explicit FD-coloring Jacobian (Jacobian-free explicitly deferred to future work)	ADODG Case 3 wing, structured hex	102,912	unstated ; “decompose”/ “processor”/“scotch” absent from the paper	avg <0.1%, worst flow row 0.195%
PAS 2019	Jacobian-free reverse AD (operator overloading) — the v2+/v5 architecture family	ADODG Case 3 wing, structured hex	102,912	serial. np=1, stated twice, by design	10-digit agreement
PAS 2019	Jacobian-free source- transformation AD — <i>ADflow</i> , not <i>DAFoam</i>	CRM wing / wing-body-tail	3.6M / 10.4M	block-based, fixed (48 / 96 cores)	11 / 6 digits

Table 10: Every gradient-accuracy verification in the method-paper corpus.

Three facts fall straight out of Table 10.

1. **No paper ever varies the decomposition of anything.** Decomposition is not an experimental axis anywhere in the corpus — not at fixed process count, not across process counts. The two papers that verify DAFoam’s v1 operator in (implied) parallel never state what the decomposition was.
2. **The operator our defect lives in has exactly one published accuracy measurement, and it is serial.** The survey states the reason in its §5.1 — running on one CPU core “allows us to isolate the impact of parallel communication on the performance” — and confirms the gap in its conclusions: “*We do not have scalability data for the Jacobian-free (operator overloading) and the full-code AD (operator overloading) options because we run the adjoint computation only in serial in the ADODG Case 3 (Sec. 5.1).*”

3. **No verification anywhere combines a refinement-interface mesh with a stated graph partitioning.** The one snappy-mesh verification — C&F 2018’s Ahmed body, our defect’s own geometry family — is of the v1 explicitly FD-assembled Jacobian, an architecture in which a tape-recording defect cannot exist, at an unstated decomposition.

The corpus checks the derivative class this defect spares. The serial variant of Section 3 looks at first like it should embarrass point 2 rather than support it: if the limiter’s tape can be 92.8% wrong in a single process, why did the corpus’s one *serial* verification of the shipped operator family return ten-digit agreement? Two answers, both readable straight off Table 10, and together they are the sharpest statement of the gap this section can make.

First, the corpus’s parallel evidence is entirely about the other architecture. The only two verifications run in (implied) parallel — and the only one run on a refinement-interface mesh, C&F 2018’s Ahmed body, our defect’s own geometry family — measure the v1 explicitly FD-assembled Jacobian. A finite-difference column re-evaluates the residual, branches and all: branch *selection* is differenced there, not recorded. A mis-taped branch cannot exist in that architecture. Every parallel accuracy number in the corpus is a measurement of the derivative class this defect spares, taken in the configuration where it could not appear.

Second, the one measurement that does exercise the recorded tape is a single case at a single converged state, and our own arms show why that cannot generalise for a data-dependent branch. The same `cellLimited/linearUpwind` token pair reads 92.8% at `np=1` on the structured NACA0012 and 0.34% at `np=1` on the Ahmed body. A branch that is not activated asymmetrically at the state you verify at is a branch you verify past. Ten digits on one structured wing certify that wing’s tape at that state — which is precisely what the survey claims and no more.

A published claim, corrected against ourselves. An earlier draft of our own upstream report asserted that “the toolchain’s journal paper reports average adjoint derivative error under 0.1% at up to 1536 cores” and that our measurement therefore contradicts a peer-reviewed number. Reading the paper in full showed that sentence to be *wrong twice*. The abstract joins two disjoint experiments with an “and”: the 1536 cores is a runtime-only table on a 10.1M-cell structured mesh; the <0.1% is a separate ~103k-cell accuracy study at an unstated processor count. **No accuracy number at any stated core count exists in that paper.** And both <0.1% figures attach to the v1 explicit-FD-Jacobian architecture — the same paper pairs them with “Implementing the Jacobian-free approach in DAFoam ... will be conducted in the future work.” The 0.1% scale is the FD-partials error floor, not a parallel-consistency scale.

This *strengthens* the case rather than weakening it: the report does not contradict a published measurement, it fills a hole the survey’s own conclusions declare.

No archival method paper covers the shipped architecture. The DAFoam publications page lists no framework paper after 2020; the `mdolab/dafoam` README’s citation section names exactly the two He et al. papers, tied to no version; a JOSS search finds nothing; Zenodo records 15636000 (v4.0.2) and 20045081 (v5.0.0) are software deposits, not papers. The v2.2 release note that shipped the Jacobian-free adjoint cites no paper for it either. *The v4/v5 rewrite we measured has no archival method paper at all*; the citation trail for the shipped matrix-free adjoint ends at papers describing other architectures.

Was the defect class disclosed anywhere? Two quotes were fetched for exactly this question, and neither is a tape caveat. C&F 2018 §2.9 names interprocessor boundary-patch state updates as “*essential for accurately computing the adjoint derivative*” — for the v1 FD path, where they are handled by executing `U.correctBoundaryConditions()` per perturbation. The v5 tape must *record* that same coupling, and its interface-cell rows are exactly where our cross-residual concentrates. PAS 2019 §A.4 acknowledges that “*some code structures remain a challenge to*

differentiate, including pointers and MPI calls”, and that ADflow’s parallel derivative correctness rests on manual assembly using properties “that the AD code cannot, in general, safely assume” — a statement about AD-across-MPI as a hazard class, made about ADflow’s development choices.

No paper states a halo-exchange tape caveat or a partition-consistency assumption for DAfoam’s operator-overloading path. There is no sentence a maintainer could point to as prior disclosure. What the quotes support is gentler and more useful: **the defect sits in a subsystem whose verification the papers explicitly scoped out, in a region their own earlier work flagged as accuracy-critical.**

4.2 The community record

A read-only novelty sweep ran **63 recorded searches across 10 venues** on 2026-08-04, with roughly 25 further pages read in full: the mdolab/dafoam issue tracker and Discussions, mdolab/id-warp issues and discussions, mdolab/pyofm, mdolab/MACH-Aero, mdolab/adflow, OpenMDAO/OpenMDAO, GitHub global search, web and scholar search, and the PETSc lists (LIAISON_NOVELTY_SWEEP_decomposition_defect.md, 3C74DC03). Nothing was posted; every URL cited in that memo was fetched in-session, and quotes are marked verbatim or via-summary.

Verdict: no prior report. No one has ever reported a DAfoam gradient that is converged and wrong under one decomposition and correct under another; any **scotch-versus-simple** dependence; any transpose-operator inconsistency; or anything of this class on a plain wall-bounded case without coupled patches. The word “**scotch**” **appears in zero issues and zero discussions in the project’s history**, and `getRotationMatrix3d` appears in zero issues or pull requests anywhere on GitHub.

What the record *does* contain makes the case sharper, not weaker:

thread	what it establishes
#946 (2026-02)	The strongest near-miss found anywhere. A user’s <code>check_totals</code> shows 93.92% analytic-vs-FD error on the PeriodicHill case; the maintainer replies, verbatim: “ <i>The PH case’ derivatives are not accurate when running in parallel. You have to run it in serial.</i> ” No mechanism offered, scoped to the periodic-boundary case
#379 (2022-12)	The same acknowledgement four years earlier — the periodic hill case “is known to have issues when running in parallel (because it has periodic boundaries; we haven’t gotten a chance to debug it yet)”, while other cases “can run in parallel without an issue”
#972 (2026-04)	Nearest <i>mechanistic</i> relative: cyclicAMI adjoint false-converges then produces 1e11–1e15 sensitivities; remedy is to force all AMI faces onto one rank. A coupled interface whose parallel adjoint treatment is wrong — but AMI patches, v4 assembly path, and garbage output rather than a cleanly converged 9%-wrong gradient
#885 (2025-09), #952 (2026-03)	Upstream’s clearest acknowledgement that decomposition affects the adjoint — but strictly its <i>convergence rate</i> . KAHIP partitioning improves the preconditioner; kahip support was added upstream 2025-12. Nobody in either thread checks the converged gradient <i>value</i> against decomposition
#101/#102 (2021-03)	The one time serial-vs-parallel gradient equality was actually checked upstream, it passed: the reported discrepancy was cell ordering in a written field, not the gradient. v1-era topology case
#914 (2025-11)	Documents upstream’s acceptance threshold: a 1.27% analytic-vs-FD deviation draws “the gradient accuracy is very acceptable for practical aerodynamic optimization”

Table 11: The community record on parallel adjoint correctness in DAfoam.

Three consequences for how the finding should be presented upstream.

1. **The “known issue with coupled patches” reply is pre-answered.** Every upstream acknowledgement of parallel-wrong derivatives attributes the wrongness to a coupled patch (periodic, AMI), prescribes avoidance, and names no mechanism. Our Ahmed-body case has *no* coupled patches: it shows the same class of wrongness arising from ordinary processor boundaries — precisely the configuration upstream’s own lore calls safe.
2. **The “decomposition only affects convergence” reply is pre-answered.** #885/#952 concern convergence rate; our KSP *converges* (reason 2, true residual 1.7×10^{-7}) to a wrong solution, which is exactly what the cross-residual measurement discriminates.
3. **The triggering configuration is the default.** The docs make no parallel-consistency claim and give no decomposition guidance at all, while the shipped default is `scotch` (pyDAFoam.py lines 590–591, main at E77F0C0C, fetched raw 2026-08-04: `self.decomposeParDict = {"method": "scotch", ...}`).

The remaining gaps are not novelty gaps. They are, unchanged: a reproducer on a stock upstream tutorial rather than a lab-built case, and mechanism to a line.

One novelty gap that is new, and is stated rather than assumed away. The sweep above ran on 2026-08-04 and was aimed at the decomposition *phenomenon*: whether anyone had reported a DAFoam gradient that is converged and wrong under one decomposition and correct under another. The limiter result post-dates it by a day and the serial variant by three, and **neither has been through a novelty sweep of its own**. Nothing here claims priority for “the slope limiter’s reverse tape is wrong”; what is claimed about it is measured, not surveyed. Closing that gap is cheap and belongs before any filing — the more so because the serial variant is the entry point a maintainer would actually run.

5 The IDWarp rotation guard

This defect is independent of Section 2 in code and in provenance — it is geometric, serial, and lives in a different repository — but it was found in the same arc, it contaminated the same gradients, and it had to be removed before the decomposition measurements could be trusted. It is not independent *in kind*: a guard is a branch, and Section 2.9 arrives at the same object from the other end. It is also the cleanest member of the class: root cause named to a line, confirmed by repair.

5.1 The line

IDWarp **2.6.2** is the exact version in `dafoam/opt-packages:latest` (image digest `sha256:9d45679d55fd47f5ca7afd99cabb86c7c2729cf2acf34c438eb33af5290f07fc`); the installed `idwarp/*.py` files are md5-identical to the `v2.6.2` tag of <https://github.com/mdolab/idwarp>, tag commit 647FD8FC.

The warp displaces each volume node by an inverse-distance weighted sum in which the weights W_i are *constant* with respect to the surface coordinates; all the surface dependence lives in a rotation matrix M_i and a displacement b_i , with $S_i = M_i \cdot (r - X_{u0,i}) + (X_{u,i} - r)$. **The rotation acts on the lever arm from the surface node to the volume node.** M_i is built at `src/modules/kd_tree.F90:1639–1640` as $R(n_0 \rightarrow n)$, by:

```
real(kind=realType), parameter :: tol = 1.4901161193847656e-08    ! line 44 (= sqrt(eps))
call cross_product_3d(v1, v2, axis)
call getMag(axis, axisMag)
! When axisMag is less than sqrt(eps), the acos 'arg' value will be
```

```

! exactly one which will give a nan in complex mode.
if (axisMag < tol) then                                ! line 58   <<< HERE
    angle = zero
    axis = zero
    axis(1) = one
else
    axis = axis / axisMag
    vv1 = v1 / magv1 ;   vv2 = v2 / magv2
    arg  = min(one, vv1(1)*vv2(1) + vv1(2)*vv2(2) + vv1(3)*vv2(3))
    angle = acos(arg)
end if
Mi = Mi + sin(angle) * A + (one - cos(angle)) * C      ! Rodrigues

```

The map $M_i(n_0, n)$ is perfectly smooth at $n = n_0$, with the finite, non-zero derivative $dM_i = [n_0 \times dn]_\times$. But *this parameterisation* — normalise the cross product, take `acos` of the dot product, apply Rodrigues — has a removable coordinate singularity there: `axis` is 0/0 and $d(\arccos)/d(\arg) \rightarrow \infty$. The guard exists to stop that becoming a NaN, and the source comment says so. It replaces the singularity with `angle = 0`.

What Tapenade does with the guard. In the generated reverse file `src/adjoint/outputReverse/vectorUtils_b.f90:123-128`, when the guard branch was taken the reverse sweep *zeroes every adjoint on the path back to v2*; the subsequent `GETMAG_B` and `CROSS_PRODUCT_3D_B` calls therefore propagate exactly zero, and $dM_i/d(\text{normals}) = 0$ exactly. Tapenade is not at fault: it differentiated the branch that was taken. The defect is that the branch taken is a *constant* where the true function has a non-zero first-order term.

The guard fires with certainty at every gradient evaluation. `getMag` regularises with $+10^{-30}$ under the square root. At an undeformed baseline `normals == normals0` bit-for-bit (same code, same inputs), so `axis` = $n_0 \times n = 0$ and `axisMag` = $\sqrt{10^{-30}} = 10^{-15} < 1.4901 \times 10^{-8}$. **Branch 0, guaranteed, at every non-corner surface node.** A `check_totals`, a first design iteration, and every gradient probe are evaluated at exactly that state. The finite difference, meanwhile, perturbs the surface by h and tilts adjacent faces by $\theta \sim h/L$, far above the threshold — so the FD sees a first-order rotation term the AD has set identically to zero. *That difference is the gap.*

5.2 Narrowing the claim, against ourselves

The obvious statement — “the rotation term is mis-differentiated” — is too broad, and measurement refuted it. Across five of IDWarp’s own regression meshes, run with the project’s own displacements:

mesh	deformation	worst err, rotations ON	worst err, rotations OFF	$\ dX_s\ $ on vs off
symm_block (inflate_cube)	axis scaling	212.6%	1.06e-05%	identical
o_mesh	shear sweep	1.7e-05%	5.8e-03%	6.42e+03 / 9.46e+02
co_mesh	shear sweep	3.9e-05%	5.1e-05%	7.27e+04 / 2.16e+03
sym_mesh	shear sweep	6.9e-06%	9.1e-04%	8.85e+04 / 2.45e+03
onera_m6	shear sweep	1.26%	9.7e-07%	1.69e+04 / 3.06e+03

On the three shear cases the deformation genuinely rotates normals, the guard does *not* fire, the AD *does* change when rotations are switched off — and with rotations **on** those cases are clean at $10^{-5}\%$ to $10^{-6}\%$. **When the rotation branch is live, warpDeriv is correct.** The catastrophic failure is confined to $n \approx n_0$. This is a sharper and more falsifiable statement than the one we started with, and it is the correct one.

The capstone that could have refuted it. If the failure is the degenerate branch specifically, then moving a case off its undeformed baseline should make the guard stop firing, at which point `warpDeriv` must become sensitive to `useRotations` and get dramatically more accurate. With all 27 shape design variables pre-set to 0.02 on the U-bend case, the two sign flips vanish and the worst error falls from 207.0% to 8.2% with nothing else changed — and the fingerprint confirms the branch is now live: $\|dX_s\|$ is $1.495168856286067 \times 10^3$ at the undeformed baseline whether rotations are on or off (*identical* — branch dead), and $1.566420804338303 \times 10^3$ vs $1.495168856286067 \times 10^3$ pre-deformed (*differs* — branch live). Had the mechanism been “rotations are mis-differentiated”, this experiment would have failed flatly.

5.3 Why upstream’s CI cannot see it

Four independent mechanisms, each verified against primary source.

1. **The assertion is missing.** `tests/test_USMesh.py:86` calls `mesh.verifyWarpDeriv(...)` and *discards the result*. Nothing is asserted on it — which is exactly what upstream issue #57 says.
2. **The only thing asserted is blind by construction.** The regression check is `handler.root_add_val("... Sum of dxs:", val, tol=1e-8)`. Across five meshes the rotation term contributes *exactly zero* to that sum (identical to 0–1 ulp) while changing $\|dX_s\|$ by factors of 5 to 34. Five for five, upstream’s check cannot detect a change that alters the derivative vector by a factor of 34.
3. **A dot-product test cannot detect it either, and is not run.** `examples/structured/-dotprod.py` lives in `examples/`, not `tests/`. And a dot-product test only proves forward-AD and reverse-AD are mutual transposes: if both linearise a *different* function than the warp, they do so identically and the dot product is still zero.
4. **The verification tool defaults to a random seed.** `UnstructuredMesh.py:460: def verifyWarpDeriv(self, dXv=None, ..., randomSeed=314)`. A random direction contracts the error against a direction it barely overlaps (0.32–1.30% where the real objective seed reads 207%, flipped).

Not a documented approximation. No documentation, release note, docstring, or source comment qualifies `warpDeriv` as approximate. The docs say only “Compute the warping derivative $(dX_v/dX_s^T) \cdot \text{Vec}$ ”; the `warpDeriv.F90` header says only “Compute the following matrix vector product: $dX_v/dX_s^T \cdot X_{vbar}$ ”. Both present it as the exact transpose Jacobian. *One disclosed gap:* the primary IDWarp reference (Secco, Kenway, He, Mader, Martins, *AIAA J.* 59(4):1151–1168, 2021, doi:10.2514/1.J059491) could not be retrieved — five umich URLs return 403 including the authors’ own bibliography link, and OpenAlex (`is_oa: false`, `oa_status: "closed"`) and Semantic Scholar independently report no open copy. No claim is made about what that paper says, and the “undocumented” characterisation rests on code, docs and release notes only.

5.4 The upstream issue this answers

`mdolab/idwarp#57`, “`inflate_cube` test appears to fail”, opened 2021-07-14, labelled bug, **open, zero comments, five years untouched**. Its timeline contains exactly one event: the label, applied 8 seconds after creation. Its author wrote: “*the output from this test appears to show large errors in some of the derivative terms. However, these error values are not actually used as a unit test and so the entire test suite still passes*”, and “*Given how weird the FD values are for the two terms with big errors, it may be that the issue is in those rather than the AD values, but it’s hard to see how that could be the case.*”

We reproduced the issue exactly — IDWarp alone, no DA Foam, no OpenFOAM, no CFD, using IDWarp’s own test mesh, options, displacement and verification routine — and answered its open question adversarially. A step-size study on the two failing degrees of freedom gives FD values stable to six significant figures across three decades ($h = 10^{-4}$ to 10^{-6}), with subtractive roundoff appearing only at 10^{-8} . **The finite difference is converged; the AD is the wrong one.**

Why degrees of freedom 0 and 3 and not 1, 2, 4, 5: `inflate_cube` scales an axis-aligned block by (1.1, 1.2, 1.3), which leaves every face normal *exactly* unchanged, so the guard fires even at the deformed state. Perturbing a node *along* its face normal tilts adjacent faces at first order and switches the rotation term on; perturbing it *in* the face plane does not. That is a clean instance of our own predicted-clean control, appearing unbidden inside upstream’s five-year-old data.

5.5 The patch, and what it proves

The derivation was written *before* the code, and every sign was checked by a controlled experiment with a hand-computed known answer before any system-level run. With $u_1 = v_1/|v_1|$, $u_2 = v_2/|v_2|$: $\sin \theta [k]_{\times} = [u_1 \times u_2]_{\times}$ exactly, and $(1 - \cos \theta)[k]_{\times}^2 = O(\theta^2)$, so $M_i = I + [u_1 \times u_2]_{\times} + O(\theta^2)$ and, at $u_2 = u_1$,

$$\begin{aligned} dM_i &= [w]_{\times}, \quad w = \frac{v_1 \times dv_2}{|v_1| |v_2|} \quad (\text{forward}) \\ v_{2b} &+= \frac{\text{axial}(m_{ib} - m_{ib}^T) \times v_1}{|v_1| |v_2|} \quad (\text{reverse, by the cyclic triple product}) \end{aligned}$$

$|v_1|$ and $|v_2|$ are already recomputed in the reverse routine’s forward sweep and are in scope inside the branch. **The whole fix is four lines of assignment in the branch .EQ. 0 block**, replacing nothing: the three zeroed intermediates stay zeroed, because the dummy-axis path they belong to really does carry zero; the patch adds the term that path lost. The diff is 2 files, +44 lines, all in `src/adjoint/output{Reverse,Forward}/vectorUtils_{b,d}.f90`.

acceptance test (stated before the runs)	before	after	verdict
Upstream <code>inflate_cube</code> (#57), rotations ON, DOFs 0 / 3	210.16% / 212.62%	8.56e-06% / 3.35e-05%	PASS
A1 NACA0012, real dCD/dXv seed, idx6 (leading-edge combo)	634.0%, sign flipped	5.54e-04%	PASS
A5 U-bend, pressure-loss real seed, idx8 / idx17	207.0% / 121.6%, both flipped	3.0e-06 / 7.0e-06 (relative)	PASS
Regression control: <code>o_mesh/co_mesh/sym_mesh/onera_m6</code> , guard dead	—	bit-identical verifyWarpDeriv output, every printed digit	PASS
Primal invariance: 310,284-coordinate warped grid	—	md5-identical (8fafe12f848a f490a5041c865112b5fb), max diff = 0.0	PASS

Table 12: Acceptance results. Note the *direction*: on #57 the AD moved to the already-converged FD ($2.87055669475 \rightarrow -115.875779229$ against an unchanged FD of -115.875789153) — the direction a genuine fix must move and a comparison-zeroing artifact could not produce. The in-plane DOFs 1, 2, 4, 5 are untouched to every printed digit.

On the A2 wing the patch improves three of four `check_totals` rows by 33.9x, 53.2x and 115.1x (CD/shape $1.713791 \times 10^{-2} \rightarrow 5.059114 \times 10^{-4}$; CL/shape $1.165163 \times 10^{-2} \rightarrow 2.189473 \times 10^{-4}$; CL/twist $1.120617 \times 10^{-2} \rightarrow 9.737824 \times 10^{-5}$) — and *degrades* the fourth, CD/twist, from 0.389%

to 0.505%, both deep inside the pass band. Any one-line summary quoting “34–115x” must carry that footnote.

Two limits, stated plainly. First, **retuning the tolerance cannot fix this.** `tol` is exactly $\sqrt{\varepsilon}$, but the `acos(min(1, v1.v2))` construction is numerically flat on its own out to $\sqrt{2\varepsilon} = 2.1073 \times 10^{-8}$ rad: $1 - v_1 \cdot v_2 = \theta^2/2$ drops below ε , `arg` rounds to bit-exact 1.0, and `acos(1.0)` is exactly 0, guard or no guard. Measured at $\theta = 10^{-8}$: `arg` bit-exactly 1.0, $1 - \text{arg}$ bit-exactly 0.0. *The first thing a maintainer will reach for is foreclosed.* Second, **the AD’s zero is the exact derivative of the implemented function at that point.** The guard makes the warp genuinely flat in a ball of angular radius $\sim 1.5 \times 10^{-8}$ rad, and a finite difference taken entirely inside that ball returns exactly 0.0, agreeing with the AD to every digit. Tapenade is correct about the code as written. The defect is precisely that `warpDeriv` returns *the derivative of a flat spot that no optimizer can traverse and no finite difference can resolve* — reaching it needs $h < 1.5 \times 10^{-8}L$, and the step study already shows roundoff destroying the FD at $h = 10^{-8}$. Any filing should pre-empt the response “the code is differentiable and your step is too coarse.”

5.6 The second regime, and the fix that covers both

Away from the baseline, with `axisMag` just above `tol`, the guard does not fire and the derivative *is* computed — but through an `acos` whose argument has lost its information to cancellation. This regime is real, measured on ONERA M6 at $\sim 1.26\%$, and **survives the patch, exactly as predicted** before the acceptance runs. Its cause was *tested* rather than asserted, by repairing each candidate line in isolation:

θ	V0 shipped	V1 denominator only	V1b angle only	V2 both	V3 singularity-free
1e-06	6.658e-05	5.561e-05	3.661e-05	1.974e-10	1.538e-12
1e-07	3.998e-04	3.339e-04	2.199e-04	7.221e-10	1.710e-12
5e-08	1.161e-02	9.663e-03	6.436e-03	3.083e-09	2.229e-12
2e-08	5.286e-02	4.483e-02	2.801e-02	4.844e-09	1.068e-12
$\leq 1.49\text{e-}08$ (guard)	1.0	1.0	1.0	1.0	9.3e-13

Table 13: Relative error of the reverse routine against the true derivative of the smooth map. V1 is the single-line repair an earlier draft of our own record named as the carrier: it buys $1.0\times$ to $1.6\times$, i.e. it is indistinguishable from doing nothing. The pair buys 2.1×10^3 to 3.7×10^7 .

The dominant term is `angle = acos(arg)` in the *primal*: $\text{arg} = 1 - \theta^2/2$ loses the information at $\theta \sim \sqrt{\varepsilon}$ before `acos` is ever called, and the reverse inherits it. Neither line is sufficient alone and the two do not add. Only `angle = atan2(axisMag/(magv1*magv2), arg)` *together with* $\sqrt{1 - \text{arg}^2} \rightarrow \text{axisMag}/(\text{magv1} \cdot \text{magv2})$ removes the regime — and the substitute is free, because `axisMag` is already computed two lines earlier in the primal.

The recommended upstream fix is neither of our patches but the reparameterisation $R = I + [v]_{\times} + [v]_{\times}^2/(1+c)$ with $v = a \times b$, $c = a \cdot b$: a polynomial in v with the smooth coefficient $1/(1+c)$, giving I with the correct derivative at $a = b$, needing no threshold, and unable to NaN except at exact reversal — a genuine singularity of the problem rather than of the parameterisation. It is measured here at 6.3×10^{-13} to 2.2×10^{-12} across every angle tested *including inside the guard ball*, where every `acos`-based variant reads exactly 1.0. It is the only single change that covers both regimes.

Two caveats attach. A regime-2 repair **cannot** be primal-bit-identical — it changes `angle` in the forward warp (M_i moves by 1 ulp at $\theta = 1$) — so any such patch must be re-graded on the primal as well as on the gradient. And our shipped patch is a *proof of concept on the generated file*: upstream should fix the primal’s parameterisation and regenerate, not merge a hand-edit of Tapenade output.

5.7 Workaround, with its price named

`useRotations: False` in `meshOptions` restores gradient consistency to roundoff on all evidence here — it makes the warp exactly linear in the surface coordinates. It is not free: upstream documents rotation interpolation as what “will help preserve boundary layer orthogonality” under large orientation changes, so mesh quality under large deformation is the price. **That trade has not been measured by this lab**, and no optimisation has been re-run under it. Recommending it as a default would be premature.

6 The conditioning wall, and the sub-LU remedy

6.1 The wall

On a 21,000-cell curved-backward-facing-step (CBFS) field-inversion case, the DAfoam adjoint linear solve fails at iteration 0 with `PetscConvergedReason: -9` (`DIVERGED_NANORINF`) and printed iteration-0 residual $7.091590452305 \times 10^{-4}$. It fails identically under all five available `jacMatReOrdering` values — two produce `-9`, three produce `-3` flat stagnation, none converges — and the printed initial residual is identical to 13 digits across all of them, which is the tell that the failure is in the factorization, not the Krylov space.

6.2 The offline reproducer, which made every candidate a ten-second experiment

The dumped CBFS system (`dRdWTPC`, 210,592², nnz 13,710,468, plus the right-hand side whose norm matches the printed iteration-0 residual to 13 digits) was solved with a `petsc4py` harness that replicates `DALinearEqn::createMLRKSP`’s call sequence *in order* — `KSPSetFromOptions` first, hard-coded GMRES/ASM/sub-ILU settings second — under `mpirun -np 4`, the real layout:

```
RESULT reason -9 its 0 finalres 7.091590452305e-04 wall 7.01s
```

Bit-for-bit DAfoam’s failure — same reason, same iteration count, same residual — with `PCView` confirming full parity (ILU fill 1, ordering `rcm`, shift `NONZERO`, pivot-in-blocks, `zeropivot 2.22e-14`).

6.3 Runtime options: reachable, measured, and empty

A standing belief in three independent internal readings held that no PETSc runtime option can reach DAfoam’s ASM sub-block factorization. That was half wrong, and the half matters. PETSc’s `PCSetUp_ASM` calls `KSPSetFromOptions` on each sub-KSP it creates *inside* `KSPSetUp` — after DAfoam’s one visible call site, before DAfoam’s hard-coded per-block overrides. Measured, not argued:

variant (all via <code>PETSC_OPTIONS</code>)	deployed-factor evidence	result
control (parity with DAfoam)	<code>zeropivot 2.22e-14</code> , <code>rcm</code> , shift <code>NONZERO</code>	-9 , iter 0
<code>-sub_pc_factor_nonzeros_along_diagonal</code>	diagonal already has no zeros	-9 , iter 0
<code>-sub_pc_factor_zeropivot 1e-8</code>	“tolerance for zero pivot 1e-08” in <code>PCView</code> — <i>the option reached the factor</i>	-9 , iter 0
<code>-sub_pc_factor_diagonal_fill</code>	applied	-9 , iter 0

Two conclusions, one per column. **Reachability:** factor options do land, so “compiled-in, nothing reachable” was too strong. **Mechanism:** a 10^{-8} zero-pivot threshold with `MAT_SHIFT_NONZERO` active still produces the NaN, so this is *not* a small-pivot-below-threshold event the shift machinery could catch. The failure is factor growth — enormous elements in L/U that overflow when applied

— which no shift and no diagonal permutation fixes; only pivoting or completeness does. This also explains, mechanically, the earlier observation that `spilu` reported “exactly singular” at every drop tolerance on the same dumped system while `splu` succeeded at every pivot threshold.

6.4 The one-word change, offline and then in-solver

sub-PC	ordering	reason	iterations	wall	per-block factor size
ILU(1) (DAFoam)	rcm	-9	0	7.0 s	6,963,844 nnz (before NaN)
LU	rcm	2 (converged)	347	68.0 s	100,311,134 nnz (~0.8 GB)
LU	nd	did not finish	—	killed at 600 s	—

Table 14: Offline mechanism test on the dumped system: one change, `PCSetType(subpc, PCLU)` where DAFoam hard-codes PCILU. The `nd` row is an operational finding worth keeping: PETSc’s native LU under nested dissection is pathologically slow on these blocks, so `rcm` is the ordering that works with complete LU here — the exact reverse of its role under ILU, where `rcm` is the ordering that surfaces the NaN.

`PCSetType(MLRsubpc, PCILU)` is hard-coded at `DALinearEqn.C:266-267`, so complete LU needs a recompile. The patch adds an environment-variable switch at that line: when `DAFOAM_SUBPC_TYPE=lu` is set, the local PC type becomes PCLU; otherwise not one line of behaviour changes. In-solver, on B3’s exact `-9` configuration:

```
DAFOAM_SUBPC_TYPE=lu: ASM sub-block PC set to complete LU
Main iteration 0    KSP Residual norm 7.091590452305e-04
Main iteration 400  KSP Residual norm 6.488778512831e-04
Main iteration 500  KSP Residual norm 1.778417281485e-05
Main iteration 667  KSP Residual norm 6.922418564747e-10
**Completed**! Total iterations: 667. PetscConvergedReason: 2. 232.2 s
```

The residual history is worth reading rather than skipping: the first 400 iterations crawl, because the assembled `dRdWTPC` the LU inverts is not the matrix-free operator GMRES applies; then the Krylov space catches the dominant cluster and the residual falls six decades in 267 iterations. Slow-then-superlinear is what a correct strong preconditioner on a mismatched operator looks like. Flat-to-thirteen-digits is what the dead ILU looked like. They are not the same thing.

6.5 The gradient, and its verification

The adjoint linear system depends on the objective, not on the design variable, so the converged solve carries over directly to the field-inversion design variable: 21,000 per-cell β components, same 667 iterations, same reason 2, gradient norm $1.4558046603 \times 10^{-5}$. **This is the first converged field-inversion gradient on a closure-relevant case in this lab.**

6.6 Where the remedy comes from, and why it is not exotic

Switching failing ILU sub-blocks to LU is the PETSc developers’ own canonical escalation, not an invention of this lab. From the `petsc-users` archive, verbatim: “*The default preconditioner with ILU(0) on each process is not appropriate for your problem and is producing overflow. Try -sub_pc_type lu and see if that produces a different result*” (Barry Smith, `msg24474`, 2015-03-26); and `msg26973` (Hong, 2015-10-27) demonstrates on a zero-diagonal system that ILU with any shift “does not converge” while `-sub_pc_type lu -sub_pc_factor_shift_type nonzero` converges in 24 iterations. The PETSc FAQ states, verbatim: “*The convergence of many of the preconditioners in PETSc including the default parallel preconditioner block Jacobi depends on the number of processes.*”

cell	h	central FD	adjoint $g[i]$	rel. err
5491 (largest $ g $)	0.05	1.914384790951e-06	1.916018813330e-06	0.085%
5491	0.1	1.907246786675e-06	1.916018813330e-06	0.460%
6740	0.05	1.679642395377e-06	1.678653382471e-06	0.059%
12486	0.05	1.472401682783e-06	1.469472928907e-06	0.199%
6490 (<i>chosen by the verification sweep, never published by the lab</i>)	0.05	1.520027166028e-06	1.519705840939e-06	0.0211%

Table 15: Central-difference verification of the 21,000-component gradient, fresh process per point, cold reset of the processor directories. The unperturbed control run reproduces the adjoint run’s objective to *exactly zero difference* ($1.5279278906359758 \times 10^{-2}$, all 17 digits) — which doubles as a primal-invariance proof across the switch. The doubling of the error from $h = 0.05$ to $h = 0.1$ on cell 5491, with the FD moving *away* from the adjoint value as the step grows, is central-difference truncation behaving exactly as $O(h^2)$ predicts: the agreement is step-limited, not noise-limited.

The complete upstream DAfoam remedy ladder for a non-converging adjoint, assembled from every place it is stated (the run-script guide, the FAQ, the doxygen DAOPTION page, and the live discussion threads) is: `renumberMesh -overwrite; pcFillLevel 1 → 2; jacMatReOrdering rcm → nd/natural; gmresRestart/gmresMaxIters up; asmOverlap up; upwind schemes; (since 2025-12) kahip decomposition; and transonicPCOption` for transonic cases. **Nothing in that ladder changes the sub-PC type; the ILU choice itself is never questioned upstream**, and no upstream doc, issue or discussion mentions ILU singularity, zero pivots, or a direct/sub-direct factorization at all — under 8 targeted searches plus 4 doc-page reads. Meanwhile `mdolab/dafoam` discussion #1002, converted to issue #1011 on 2026-07-29 and still open, shows users exhausting the documented ladder on *official tutorials* and stalling: the failure mode this switch was built for is live upstream today.

The shape of a filing. An upstream pull request exposing the sub-PC type as an `adjEqnOption` key — default `ilu`, behaviour unchanged — would be novel, is one option plumb, and is backed by both a measured unblock and named-PETSc-developer precedent. Our local implementation is one hunk at `DALinearEqn.C:266` (`demo-output/website/dafoam/subpclu_patch/DALinearEqn_subpclu.patch`, regenerated at 9796F90F); its default-off behaviour was verified by re-running the unpatched configuration on the patched image and reproducing the `-9` at iteration 0 with all 13 residual digits intact. *Added 2026-08-07:* the committed diff is now mechanically applicable. The version this document originally cited carried absolute headers into a scratch directory that no longer exists, so `git apply` failed on the very file cited as the reproduction diff — filed by a family supervision pass and fixed by regenerating against the stock `DALinearEqn.C` extracted from the shipped image with `a/src/adjoint/DALinearEqn/DALinearEqn.C` headers. **`git apply -check now` PASSES** from the DAfoam repository root, and the 19-line applied block was verified byte-identical to the change in the deployed `dafoam-subpclu:v1` image. One declared delta: the regenerated patch adds a warning on a set-but-unrecognized `DAFOAM_SUBPC_TYPE` value (`LU`, `Lu`, a trailing space — all of which previously fell through to stock ILU in silence), and the `v1` image is *not* rebuilt, so that warning exists in patch text only and drivers must keep asserting the sub-LU log line until a rebuild lands.

6.7 What the change does *not* fix

On the NASA hump (51,626 cells, 51,626 β design variables) the same one-word change moves the failure from catastrophic to honest, and no further. The `-9` is gone — the complete-LU factorization of blocks ILU cannot factor completes, and GMRES iterates with a monotonically descending residual, 1.094138002900 at iteration 0 to 0.9544468674795 at iteration 900 — but it descends slowly, 12.8% in 900 iterations (accelerating: 0.75%, 1.1%, 1.2%, 1.6% per successive

hundred) and far from the target. The run was stopped deliberately at roughly iteration 900 when host available memory fell from 30.5 GB to a low-water **1.62 GB** under about 19 GB of LU factors plus a still-filling 2000-vector restart space, on a shared box.

Stated for whoever picks this up: **the hump blocker is no longer the singular factorization — it is Krylov convergence rate against a memory envelope**, which is a different and honestly-named problem.

7 Methods: adversarial verification as practice

Every headline claim in this document has survived an independent adversarial sweep. That phrase is used precisely, and this section says what it buys, because a reader assessing four unfilled defect claims is entitled to know what standard they were held to.

7.1 What a supervisor sweep is

A verification agent is dispatched with the posture that **the claim is wrong until it survives attack**. It does not review the record; it re-derives the record. Concretely, across the six sweeps behind this document:

- **Independent instruments, not re-runs of the lab’s scripts.** The mechanism sweep rebuilt the processor-to-serial permutation from scratch with its own parsers, from `cellProcAddressing`/`faceProcAddressing` and the `AdjointIndexing` dumps, then compared against the stored map: identical bit-for-bit in target and sign. The rotation sweep wrote its own NumPy check of the corrected derivative (250 draws, against Richardson central differences of the singularity-free reference) and its own DAfoam driver that wraps `idwarp.USMesh.warpDeriv` itself rather than the lab’s hook.
- **A new validation channel the original never used.** The `AdjointIndexing` dumps carry cell and face centre coordinates; mapping `np=4` coordinates onto `np=1` coordinates gives max deviation **0.0** over all 26,149 and 26,227 entries, validating every entry of the permutation exactly and independently of any solved field.
- **Re-measurement from scratch, in the container.** One cell of the phenomenon table (`np=2`, patched) was re-run with the sweep’s own driver, own staging, own container invocation: analytic, FD, absolute error, relative error, baseline C_D and GMRES iteration count *all identical to every printed digit*, and carrying the provenance stamp the archived run lacked. Cost: 86 s wall, 2.9 solver core-min.
- **A point the lab did not choose.** The conditioning sweep verified a finite-difference cell (6490) the lab never published, at a different rank band, and got 0.0211% — tighter than any published cell. A wrong indexing, a stale gradient file, or a lucky triple would not survive that.
- **Cold reproduction of the control, not just the treatment.** The env-off regression was re-run cold by the sweep, on a fresh case copy: reason, iteration count, and all 13 residual digits matched independently of anything the original session wrote.
- **Recomputation from raw vectors, trusting no staged intermediate.** The robustness sweep recomputed both of Table 6’s cross-residuals with its own NumPy from the dumped vectors (1.354841e-02 and 1.047165e+00), rebuilt the `np4→np1` permutation from the raw addressing files — its reconstruction equals the staged input under `np.array_equal`, for both levers — verified the sign convention rather than assuming it, and confirmed *in code* that the instrument’s operator is the matrix-free AD product the KSP uses and never the assembled matrix dumped alongside it.

- **A byte-level audit of the “one word” claims, because they are load-bearing.** `diff` of the R6b case against the established source: one changed line, in `fvSchemes`, everything else `cmp-clean`. `diff -r` of the serial arm against the clean case it was built from: the single differing input file is `system/fvSchemes`, two lines; and its one-word control differs by exactly one word.
- **A free check the record never ran, found by attacking the record’s own logic.** The serial-limiter sweep judged the record’s step-stability test logically insufficient against a symmetric FD kink — correctly — and then closed the hole at zero cost by mining the one-sided slopes out of the FD legs’ own printed converged C_D values (Table 9), first validating the reconstruction by rebuilding the printed central-difference vector from the same values to 4.7×10^{-9} .
- **The ledger is audited to the second.** Billed walls were reconciled against log mtimes and internal timings — e.g. five offline runs plus a 600 s kill summing to 693.04 s against 693 billed; a memory trace whose first and last samples bracket 816 s against 819 billed; nine finite-difference runs whose per-run walls sum to *exactly* 642; and 21 robustness-campaign lines re-summing to exactly 128.28 with all 21 per-line `wall` $\times$ `cpus` arithmetic checks passing.

7.2 What the sweeps actually found

They did not rubber-stamp. Across the six sweeps the claims survived on every axis — one of them “CONFIRMED WITH A NAMED ERRATUM” — and the sweeps still returned a defect list, which is printed here rather than repaired quietly. The mechanism sweep found five bookkeeping and provenance defects; the phenomenon sweep found two documentation defects and three wording over-reads; the rotation and conditioning sweeps found no defect and two presentation caveats between them; the robustness sweep found one factual error and named three protocol gaps; the serial-limiter sweep found no error, named two, and ran the check whose absence was one of them. The material ones appear in the body of this document; the consolidated list:

7.3 Pre-registration

Predictions are committed to the repository *before* the runs that test them, and scored exactly as written.

- The cross-residual discriminator was proposed with the prediction “the cross residual will be large.” **HELD:** 3.3×10^2 against a 1.1×10^{-4} floor.
- The coloring-off discriminator was pre-registered with “the gradient will not change, because coloring builds only the preconditioner.” **NOT SCORED** — the discriminator proved unrunnable, so the prediction never met its measurement. It is scored NOT SCORED rather than quietly dropped or retroactively counted.
- The rotation root-cause session committed seven falsification conditions (P1–P7) before any diagnostic ran, including two — “`evalMode=exact` must *not* help” and “matrix reordering must not move the result at all” — whose failure would have refuted the mechanism. P5 and P6 both held; P7’s designed control came out only partially clean and is reported at its real strength, with the genuinely clean instance found elsewhere.
- The reach matrix (Section 2) registered seven arms with numeric bands and an explicit decision rule for refutation, at 99F5D41D, before any arm ran.
- The robustness campaign registered nine arms across a pre-registration and four amendments, every amendment committed *seconds* before its arms launched — the sweep reconciled each commit timestamp against ledger completion minus wall (17:14:47Z before a 17:15:08 start,

found	disposition
The cross-residual table's numbers appear in no log; the instrument carried a sign error and every arm's log prints a degenerate <code>ratio=2.000000e+00</code>	Stated in full, Section 2.4; values are exact offline corrections and reproduce three ways including a fresh serial run. <i>The instrument itself was repaired 2026-08-07</i> — and reproduces the floor to every digit with no correction at all, which makes the repair a fourth reproduction
The instrument was left unrepaired after the sweep said to repair it, and was then reused twice with two same-family filename conventions and nothing declaring which	Filed by a family supervision pass as standing and high-severity; fixed at both sites with a convention banner and a distinct post-fix output prefix. No published number moves
The committed sub-PC diff carried absolute headers into a directory that no longer exists, so <code>git apply</code> failed on the file this document cites as its reproduction diff	Regenerated with repo-relative headers, <code>git apply -check</code> PASS, applied block verified byte-identical to the deployed image
The either-copy confound bound silently used the assembled matrix — the instrument the same record discredits	Disclosed here; immaterial by a 30× margin even when inflated by the full assembled-vs-AD discrepancy
The cited coloring-crash log no longer exists (the runner truncates per attempt; three attempts shared a tag)	Claim re-established statically in the shipped container, by code rather than by the cited log
The residual-function-identity loophole was left unacknowledged	Closed from the prior record's per-run FD column, and now stated in the body as its own subsection
The case's own ladder record still asserted the superseded verdict, in both directions	Filed as a lesson (L-32); satellites had moved, the primary record had not
A misprinted relative spread (1.7e-06 for 1.7e-05) and an over-broad “any group” invariance sentence (one noise-floor component differs by 11.64%)	Corrected; neither touches a number in any claim
A cost figure of 12.63 core-min that should read 12.80, repeated in a commit message	Corrected in place, with the arithmetic shown
Two headline numbers quoted in different units in a one-line summary; an “improves 34–115x” claim that passes over a fourth row degrading 0.389% to 0.505%	Both carried into this document with the footnote attached
Provenance stamps missing from the headline sweep's logs	Mitigated by a stamped independent re-run of one cell and a stamped pair reproducing the same two numbers
The established arm's adjoint Krylov count was written 590 in six places; 590 belongs to a different arm (<code>simple</code> 2x2x1, 1.40% error), and fifteen logs of the established arm read 719	Corrected at every site, here and in the source record. The correction enlarges the claim: the conditioning collapse is 17.5x, not the 14.4x reported
The R5 verdict bands were registered in absolute multiples of $\ b\ $ and mis-calibrated by ~ 5 orders against the actual $n_p=1$ floors; by the registered wording alone, the lever that <i>hides</i> the defect counts as a collapse	Printed with the split stated (Section 2.10): the substance rests on a floor-ratio criterion that was not a registered band, and it runs against the lab's own prior framing. Future cross-residual bands go in units of the configuration's own floor
No state-override control was run for either R5 lever, so the linearization-state confound is bounded only indirectly	Named in the body with both bounds; the ~ 0.4 core-min control is unbought and is on the open list
The serial arm's step-stability check cannot, by itself, exclude a symmetric FD kink — and the closing check was already free inside the record's own logs	Run by the sweep and printed as Table 9; adopted as the registered closing move for this symptom class
The “no acquisition” verdict is scored on an instrument adopted after the arm's numbers were in hand	Labelled post-hoc in the body in so many words, per the sweep, rather than merely caveated

and three more of the same shape) and found the ordering intact every time. Its scoring runs harsh: **four NOT HELD and two NOT SCORED**, including the arm whose failure named the carrier (R3a: registered $\geq 4\%$, measured 0.228%) and one whose “direction held, magnitude under the band” is reported as NOT HELD rather than upgraded. The load-bearing clean clause, R6b, was registered together with the result that would have refuted the whole hypothesis.

- The acquisition arm registered an escape clause for a dirty control, and the clause fired: the np=1 control read 92.8%, so the arm is reported NOT SCORED against its own bands rather than re-read into a result. Its contingent arms were authorized only on a fire that did not happen, and were correctly not run.

7.4 Cost, stated including the overruns

work	spent (core-min)	budget
Decomposition discriminators (8 container runs)	56.77	45
— of which the five arms that decided the item	12.80	
— of which three attempts at the coloring-off arm, all ending in a measured failure mode rather than a number	43.97	
Mechanism supervisor sweep (one 9 s container run; everything else arithmetic on dumps)	0.30	
Phenomenon supervisor sweep (one np=2 <code>check_totals</code>)	2.90	~30
Conditioning unblock (7 entries: offline ladder, regression control, two converged solves, hump attempt, 9-point FD sweep)	185.0	240
Conditioning supervisor sweep	18.3	22
Robustness campaign (21 ledger lines: two refined-mesh arms, four scheme knobs, the conformal pair, two levers at three container tasks each)	128.28	150
— of which one arm that produced no gradient at all (refined mesh, limiter present, adjoint diverged), charged in full	33.60	
Acquisition arm and its two separating arms (4 runs, all rc=0)	15.16	30
Robustness supervisor sweep (offline: raw-vector recomputation, byte diffs, two cut re-classifications)	0.00	
Serial-limiter supervisor sweep (offline: log mining, one-sided-slope reconstruction)	0.00	

Table 16: Solver cost. The one overrun — 26% on the discriminators — is confined entirely to the coloring-off arm, and the record volunteers that the second forced attempt should not have been launched after the first showed unbounded memory growth rather than a slow finish. Every run is billed at full cap occupancy for its whole wall clock, including both kills and including the refined-mesh arm that returned a failure instead of a number. The two most recent sweeps cost *nothing*: both re-derived their verdicts from artifacts already on disk, and the check that decided the serial finding (Table 9) came out of logs the record had already written and not read.

7.5 The lessons this arc produced

The method itself is on the record, as numbered lessons in `LESSONS.md`:

- **L-29.** A guard that replaces a removable singularity with a constant differentiates to a hard zero — at exactly the point every gradient evaluation uses. Audit every such guard on a differentiated path; require the guarded branch to carry the analytic limit, or reparameterise so no guard is needed. And when *testing* derivative code, evaluate at the degenerate/baseline point deliberately and seed with the real objective direction.
- **L-30.** A gate whose threshold lies between two constraint values reports the constraint, not the quantity. Before gating on a reported extremum, find out whether the tool was *told* that number.
- **L-31.** A verification instrument must itself be verified at the component it condemns. When an analytic derivative disagrees with a finite difference, re-measure the finite difference by an independent path — with a neighbouring component as a control — before attributing the gap to the derivative.
- **L-32.** When a verdict moves, the case’s own record must move first; satellites are not the record.
- **L-33.** A correction trained inside a gated regime rebuilds the gate’s own failure at the regime’s floor.
- **L-34.** “Unreachable at runtime” is a claim about the deployed object, not about one call site. Deploy a probe that prints the final object state (for PETSc, PCView on the sub-PC after setup) before declaring a configuration layer unreachable.
- **L-35.** *A converged Krylov solve certifies the operator it was given, not the operator you meant.* When an iterative solve converges cleanly and the downstream answer is still wrong, stop tuning the solve and test the operator: apply the converged solution under an independently evaluated instance of the operator — serial where the suspect is parallel, assembled where the suspect is matrix-free — and compare the residual to that instance’s own floor. It is a mat-vec, not a solve: seconds, not core-minutes. Two footnotes: map vectors across decompositions with the integer addressing files (sign on flipped faces), never coordinate matching, and validate the map on the primal state first; and diff operator *actions* on mapped vectors, never operator *entries*.
- **L-36.** *A gradient check certifies a contraction, not an operator* — and the two do not scale together. Passing `check_totals` at one configuration certifies the contraction of the operator’s error with that one objective’s adjoint direction: almost no information about the operator, none about other objectives, other design variables, or other decompositions. The converse holds too: a large measured operator defect does not imply the published gradients are badly wrong. Never extrapolate a passing gradient check across configurations; when an operator defect is suspected, measure the operator.
- **L-37.** When a differentiated code is wrong in parallel, *suspect a recorded branch first* — limiters, switching or mixed boundary conditions, guards, `min/max/abs` — and one-knob them before theorizing about halo exchange in general, because a branch is removable by a one-line dictionary edit and is therefore the cheapest discriminator available. And any lever that turns a gradient error off must be re-verified at *operator* level before the configuration is described as unaffected: a lever can fix the operator or merely rotate the contraction away from the damage, and the gradient number cannot tell them apart. State which one you measured.

- **L-38.** *An invariance check can pass because both sides are wrong together.* Decomposition-, ordering-, rank-count- and restart-invariance detect *inconsistency* between two evaluations of the same tape; a tape wrong identically in both sails through all of them. Never promote an invariance pass to a correctness verdict unless one side is anchored to an external reference. And the converse of L-37’s trigger: a recorded branch can corrupt the reverse sweep in *serial*, with the parallel machinery innocent — so when a serial gradient check fails against a step-stable finite difference, put the tape’s branches at the top of the suspect list there too, before FD-quality theories.

A Reproduction appendix

A.1 Environments

image / package	identity
dafoam/opt-packages:latest IDWarp 2.6.2 (inside the above)	DAFoam 5.0.0, OpenFOAM v2506, PETSc 3.15.5. The stock toolchain; all phenomenon and mechanism measurements digest sha256:9d45679d55fd47f5ca7afd99cabb86c7c2729cf2acf34c438eb33af5290f07fc; installed idwarp/*.py md5-identical to tag v2.6.2, tag commit 647FD8FC
patched IDWarp	scratch clone at /home/ubuntu/certonomous-runs/W5-patch/idwarp; git diff byte-identical to rotation_branch/idwarp_v2.6.2_degenerate_branch_fix.patch (2 files, +44 lines). Mounted, never installed in place
dafoam-subpclu:v1	local-only docker commit of the stock image with one hunk at DALinearEqn.C:266 rebuilt in all three AD modes. Nothing pushed

A.2 Switches and stamps

```
# Which IDWarp is actually loaded -- stamped into every log by the driver.
IDWARP_IMPORTED_FROM: /patch/idwarp/idwarp/__init__.py      # patched arm
IDWARP_IMPORTED_FROM: <container site-packages>             # stock arm

# Sub-block preconditioner switch (patched image only; default off).
docker run -e DAFOAM_SUBPC_TYPE=lu ... dafoam-subpclu:v1
# Log line emitted on block 0 when active:
#   DAFOAM_SUBPC_TYPE=lu: ASM sub-block PC set to complete LU

# The IDWarp workaround, for anyone who needs a consistent gradient today
# (mesh-quality price NOT measured -- see Section 5.7):
meshOptions = {..., "useRotations": False, ...}
```

A.3 Varying the decomposition without falling into the trap

Editing `system/decomposeParDict` directly is silently ineffective: `pyDAFoam` regenerates it. Set the `daOption` instead, and *read the decomposition back out of the log*:

```
daOptions = {
  # ...
  "decomposeParDict": {"method": "simple", "simpleCoeffs": {"n": [4, 1, 1]}},
  # default when unset is {"method": "scotch"} -- pyDAFoam.py:590-591
}
```

```
# Verify the edit took effect -- this line is the check that catches the trap:
grep "Decomposition method" run.log
#   Decomposition method scotch [4] (region region0)
#   Decomposition method simple [4] (region region0)
```

A.4 The cross-residual protocol

Pure DAfoam public API; no rebuild required. This is the instrument any candidate fix must satisfy.

```
# 1. Per decomposition: run the primal + adjoint, then dump
#   psi   = DAfoamSolver.psi
#   state = the converged primal state vector
#   b     = dFdW (the adjoint RHS)
#   with, in daOptions:
#     "writeJacobians": ["adjointIndexing"]
#   which emits AdjointIndexing_<rank>_of_<n>.txt (indices + cell/face centres).

# 2. Build the exact integer permutation to serial ordering from
#   processor*/constant/polyMesh/cellProcAddressing
#   processor*/constant/polyMesh/faceProcAddressing (sign => flipped face)
#   NEVER by coordinate matching. Validate it on the PRIMAL state first:
#   duplicated processor-face phi copies must agree to machine precision
#   (measured here: 2.7e-15). The AdjointIndexing centres give a second,
#   exact channel: max |dxyz| must be 0.0.

# 3. Evaluate the true residual under the SERIAL operator, matrix-free:
#   r = solverAD.calcJacVecProduct(state -> residual) applied to mapped psi
#   SIGN CONVENTION: OpenMDAO seeds -dF/dW, so the system is  $A^T \psi = -b$ .
#   An instrument computing  $(A^T \psi - b)$  prints a degenerate ratio of exactly
#   2.0 for its own-operator control -- that is the tell that the sign is wrong.
#   Correct offline as  $res + 2b$ , or -- better, and done here on 2026-08-07 --
#   fix the instrument: both reused tasks now compute  $(A^T \psi + b)$ , every log
#   line prints a convention banner, and the fixed task writes a distinct
#   w4xf_res*.numpy prefix so pre-fix dumps (needing +2b) cannot be conflated
#   with true residuals. The repaired tool reproduces the np=1 floor
#   1.140697e-04 with NO offline correction, to every digit.

# 4. Judge against the SERIAL operator's own floor, not against zero:
#   np=1 control ..... 1.14e-04 of ||b|| <- the floor
#   np=4 simple 4x1x1 .... 9.37e-02
#   np=4 scotch ..... 3.288e+02 <- the defect
```

Lab scripts implementing this: W4-a4-discriminators/runScript_w4.py (tasks w4_dump, w4_crossres, w4_crossres2), build_mats.py, analyze_mats.py.

A.5 The two one-word reproducers

Both defects of the branch class in DAfoam are switched on and off by a single token in system/fvSchemes. Neither needs a rebuild, a patch, or a lab artifact. The token is the same in both.

```
// system/fvSchemes -- the switch, both cases
gradSchemes
{
    default Gauss linear; // unlimited
    limited cellLimited Gauss linear 1; // the branch: min/max over the stencil
}
divSchemes
{
```

```

    div(phi,U) bounded Gauss linearUpwind limited;    // defect ON
// div(phi,U) bounded Gauss linearUpwind default;    // defect OFF -- one word
}
// Same scheme family, same order, the same gradient-correction term still evaluated
// and still needing its halo exchange. Only the selection leaves the tape.

```

(A) Parallel — the decomposition-dependent adjoint. A4, 2,777-cell snappy-refined Ahmed body, freestreamVelocity farfield, np=4 shipped-default scotch, check_totals at step 1e-3 central:

```

limited: dCD/dshape 2.2086e-01 vs own FD 2.4258e-01 -> 8.95 % KSP 719
default: dCD/dshape 2.0572e-01 vs own FD 2.0399e-01 -> 0.849% KSP 41
# and at operator level, by the cross-residual protocol above:
limited: ||A^T psi + b|| = 3.288e+02 x ||b|| (serial floor 1.14e-04)
default: ||A^T psi + b|| = 1.355e-02 x ||b|| (serial floor 2.00e-06)
# Control, same discretization at np=1: 0.016 % -- the ~0.8 % that survives
# is an FD-column shift; the analytic reads 2.0572e-01 at BOTH decompositions.

```

(B) Serial — one process, no MPI in the loop. A1, 4,032-cell structured NACA0012 with the freestreamVelocity farfield installed (inout patch; freestreamValue uniform (9.959800351 0.895754972 0) at the case's own α_0), primalMinResTol 1e-8, np=1:

```

limited: dCD/dshape 1.066449e-01 vs own FD 6.194868e-02 -> 92.8 % (CL 7.86%)
default: dCD/dshape 6.516374e-02 vs own FD 6.522800e-02 -> 0.121% (CL 0.018%)
# One of the eight components (index 6) is sign-flipped: AN -0.02591213
# against FD +0.00849673. Krylov 95/96 vs 32. No decomposition anywhere.

```

The closing check this symptom class needs, and it is free. Before attributing a serial analytic-vs-FD gap to a limiter, rule out the finite difference — and rule it out *properly*, because a symmetric kink at the base point is invariant under a change of central-difference step and will survive a two-step check. The FD legs have already printed what is needed:

```

# Each central-difference leg prints its own converged objective. Mine them:
# forward slope = (J(+h) - J(0)) / h
# backward slope = (J(0) - J(-h)) / h
# 1. Validate the reconstruction first: rebuilding the framework's printed
# central-difference vector from the same mined values must agree (here:
# 4.7e-09 absolute, at both h = 1e-3 and 3e-3).
# 2. A kink splits the two one-sided slopes apart. A smooth function does not.
# 3. The verdict is whether the ANALYTIC lies inside [backward, forward].
# Measured here on the sign-flipped component: forward +0.009660,
# backward +0.007334 (27.4% asymmetry -- real limiter chatter), all four
# secants across both steps in [+0.0053, +0.0143], analytic -0.025912.
# Outside the bracket, wrong side of zero, 3x the magnitude: no branch
# selection available to the FD produces that. The adjoint is convicted.

```

A.6 Reproducing the IDWarp defect in isolation

No DAfoam, no OpenFOAM, no pyGeo, no CFD — IDWarp's own test mesh, options, displacement and verification routine:

```

# input_files/symm_block.cgns, md5 0d5a0411cbaa00f6afd88a9430a7f634,
# from upstream's own input_files/get-input-files.sh
mesh.verifyWarpDeriv(dXv_warp, solverVec=False, dofStart=0, dofEnd=5)
# Rotations ON : DOF 0 AD 2.87055669475 vs FD -115.875789153 -> 210.16%
# Rotations OFF: DOF 0 AD 2.87055669475 vs FD 2.87055639160 -> -1.06e-05%

```

```
# ~ the AD column is BIT-IDENTICAL between the two runs: the proof that
# the rotation term was never in the derivative.
# Adversarial check on the FD: stable to 6 significant figures over
# h = 1e-4, 1e-5, 1e-6; roundoff only at 1e-8. The FD is converged.
```

A.7 Rebuilding the sub-LU image

```
docker run -d --name dafoam-build dafoam/opt-packages:latest sleep infinity
# Apply the committed diff (one hunk at DALinearEqn.C:266, env-gated). Since
# 2026-08-07 the diff carries repo-relative a/ b/ headers and is mechanically
# applicable -- run this from the directory containing src/, in-container
# /home/dafoamuser/dafoam/repos/dafoam:
git apply --check demo-output/website/dafoam/subpclu_patch/DALinearEqn_subpclu.patch
git apply demo-output/website/dafoam/subpclu_patch/DALinearEqn_subpclu.patch
# then, INSIDE the container, for each of the three AD modes:
source /home/dafoamuser/dafoam/loadDAFoam.sh # original
cd repos/dafoam && wmakeLnInclude src/adjoint && (cd src/adjoint && wmake -j 8)
# ADR and ADF: sed WM_AD_MODE in OpenFOAM-AD/etc/bashrc, re-source, wmake again
docker commit dafoam-build dafoam-subpclu:v1
# Caution: 'set -e' before sourcing the OpenFOAM environment kills the shell.
# Source first. All three libDASolver*.so relink in seconds.
```

A.8 Artifact index

All paths are in the Certonomous repository unless marked as a run directory. Run directories live outside the repository, on the measurement host.

artifact	what it holds
Repository: demo-output/website/dafoam/	
VERIFICATION_A4_decomposition_supervisor_sweep.md (27D25762)	The verified phenomenon table, five verification axes, the hanging-node refutation
DISCRIMINATORS_A4_decomposition_mechanism.md (+ 2026-08-04 amendments)	The four discriminators, the cross-residual, the localization, the coloring-off finding
VERIFICATION_A4_mechanism_supervisor_sweep.md (8E0A08BC)	Adversarial sweep of the mechanism: independent map, independent scripts, fresh serial run, five defects found
UPSTREAM_BUG_REPORT_decomposition_adjoint.md	The prepared, unfiled report (D1). It now carries the papers-protocol correction of Section 4 in its own header, the reach addendum, a 2026-08-07 robustness addendum (“the trigger is a recorded BRANCH ... and one previously ‘clean’ configuration is wrong at operator level”), and a 2026-08-07 acquisition addendum carrying D1s. Still NOT FILED
UPSTREAM_BUG_REPORT_mesh_warpDeriv.md	The prepared, unfiled report (D2)
DEFECT_ROBUSTNESS_mesh_and_setup.md (pre-registration 25F52868, results 49631723, R7 85DFBE68)	The robustness and acquisition campaigns: the scheme ladder, the two levers at operator level, the conformal pair, the refinement arms, the seven-case survey, the serial variant, and every registered prediction scored as written

artifact	what it holds
VERIFICATION_defect_robustness_supervisor_sweep.md (8BD47B35)	Adversarial sweep of the robustness campaign: cross-residuals recomputed from raw vectors, map rebuilt bit-for-bit, the 590/719 erratum, three named protocol gaps
VERIFICATION_A1_serial_limiter_supervisor_sweep.md (0858E640)	Adversarial sweep of the serial variant: byte-level config audit, independent re-extraction, and the one-sided-slope reconstruction that closed the FD-kink hypothesis at zero cost
SUPERVISOR_FAMILY_REVIEW_2026-08-07.md	The instrument-level review the number sweeps cannot do: unit assumptions, silent fallbacks, reuse traps. Sources the sign-convention repair (A-1) and the applicable sub-PC diff (B-2), each with its OUTCOME line
DAFOAM_PAPERS_VERIFICATION_PROTOCOLS.md (BF6AC53B)	Three method papers read in full; the protocol table of Section 4
LIAISON_NOVELTY_SWEEP_decomposition_defect.md (3C74DC03)	63 searches, 10 venues, the query log, the near-misses, the PETSc precedent
ROOTCAUSE_getRotationMatrix3d.md, PATCH_getRotationMatrix3d.md	The line, the mechanism, the derivation-before-code, the acceptance results
VERIFICATION_rotation_patch_supervisor_sweep.md (6782D33A)	Adversarial sweep of D2: math re-derived, own driver, own reproduction
ladder-b/W4_ADJOINT_PC_UNBLOCK.md	The conditioning wall, the offline ladder, the sub-LU unblock, the FD gate
VERIFICATION_cbfs_unblock_supervisor_sweep.md (9C19CCC8)	Adversarial sweep of D3: image-vs-diff audit, recomputed FD table, new cell, cold regression
DEFECT_REACH_decomposition_cases.md (99F5D41D)	The pre-registered reach matrix, since scored and audited (sweep 192C970C)
subpclu_patch/DALinearEqn_subpclu.patch (1CD44C04, regenerated 9796F90F)	The one-hunk sub-PC source diff. Repo-relative headers since the regeneration; <code>git apply -check</code> passes from the DAFoam repo root
rotation_branch/idwarp_v2.6.2_degenerate_branch_fix.patch	The four-line derivative correction (2 files, +44 lines)
PROOF.md §25.5, §23–24, §25.2–25.3	The primary measurement record
LESSONS.md L-29 – L-38	The method-of-investigation thread
Run directories: /home/ubuntu/certonomous-runs/	
W4-a4-stepsweep/	The phenomenon table's per-cell logs
W4-a4-discriminators/	Per-arm dirs, dumped operators and vectors, <code>ledger.txt</code> , the analysis scripts
W4-a4-verify-audit/	The mechanism sweep's own staged case, own mapped vectors, <code>audit_crossres.log</code>
VERIFY-a4-np2-20260804/	The independent re-run of one phenomenon cell
W4-defect-reach/	The reach matrix's arms and <code>ledger.txt</code>
W4-defect-robustness/	The scheme ladder, the lever dumps and cross-residuals, the conformal and refined meshes, the A1 acquisition arms; <code>ledger.txt</code> (21 audited lines) and <code>ledger_r7.txt</code>
W5-idwarp-source/, W5-patch/, W5-rotation-branch/	IDWarp source clone, patched clone, and the A5 case copies

artifact	what it holds
W4-adjoint-pc-unblock/	Offline petsc4py ladder, CBFS and hump runs, the 9-point FD sweep
W4-verify-sweep/	The conditioning sweep's isolated case copy and its own logs

A.9 What is still open

1. **D1/D1s, mechanism to a line.** The subsystem, the spatial signature and now the recorded construct are named — the limiter's `min/max` selection, switched by one word — but the defective source line in the `cellLimited/limitedGrad` reverse sweep is not. Reaching it still requires an instrumented rebuild of `libDASolver.so`; what has changed is that the reproducer to instrument can now be a single process (Section 3).
2. **D1, reproducer on a stock upstream tutorial.** The measured case is lab-built — its mesh recipe and case directories are archived and shippable, but it is not an official tutorial. A tutorial-based reproducer varying the `decomposeParDict` `daOption` on an official case should exist before filing.
3. **D1, the one-mesh-family confound.** Not repaired — *confirmed*, twice, by the two sharpest tests available: the BC alone transplanted onto a clean case from another mesh family does nothing, and both ingredients together do nothing either. Every decomposition-borne measurement in this lab still lives on one 2,777-cell Ahmed background family. A filed reproducer must scope itself accordingly.
4. **D1, the unseparated third correlate.** `primalMinResTol` is $1e-4$ on both defective cases and $1e-6/1e-8$ on all five clean ones, and it has no within-case control, because `DAFoam` treats an unreached tolerance as a primal failure rather than a depth setting. Whether a loose primal is a necessary co-ingredient is unmeasured, and the acquisition null raised rather than lowered its standing.
5. **D1, the unbought operator control.** No state-override control was run for either lever of Table 6; the linearization-state confound is bounded indirectly and the ~ 0.4 core-min direct control should be taken next time a container is warm.
6. **D1s, novelty.** The community sweep predates the limiter findings and has not been re-run for them. No priority is claimed, and the gap should close before any filing.
7. **D2, the primary reference.** The IDWarp AIAA paper is unread and unobtainable from this host; the “undocumented” characterisation rests on code, docs and release notes only, and is stated conditionally throughout.
8. **D2, the second regime.** Real, measured at $\sim 1.26\%$ on ONERA M6, untouched by the shipped patch by design. The reparameterisation that subsumes both regimes is derived and numerically validated but not implemented, and cannot be primal-bit-identical.
9. **D2, the workaround's price.** The mesh-quality cost of `useRotations: False` under large deformation has not been measured.
10. **D3, the hump.** Krylov convergence rate against a memory envelope. A prepared configuration wrapping the exact sub-solve in Richardson with a capped restart is staged but unrun.

Filing status, restated plainly. Nothing in this document has been filed, sent, or posted — not the decomposition report, not its three addenda, not the comment on `mdolab/idwarp#57`, not the sub-PC pull request. All of it is **PREPARED and HELD**. Submissions are parked at the owner's discretion, and whether any of it goes upstream is the lab principal's decision, not the authors'. This document is what the case file looks like while it waits.